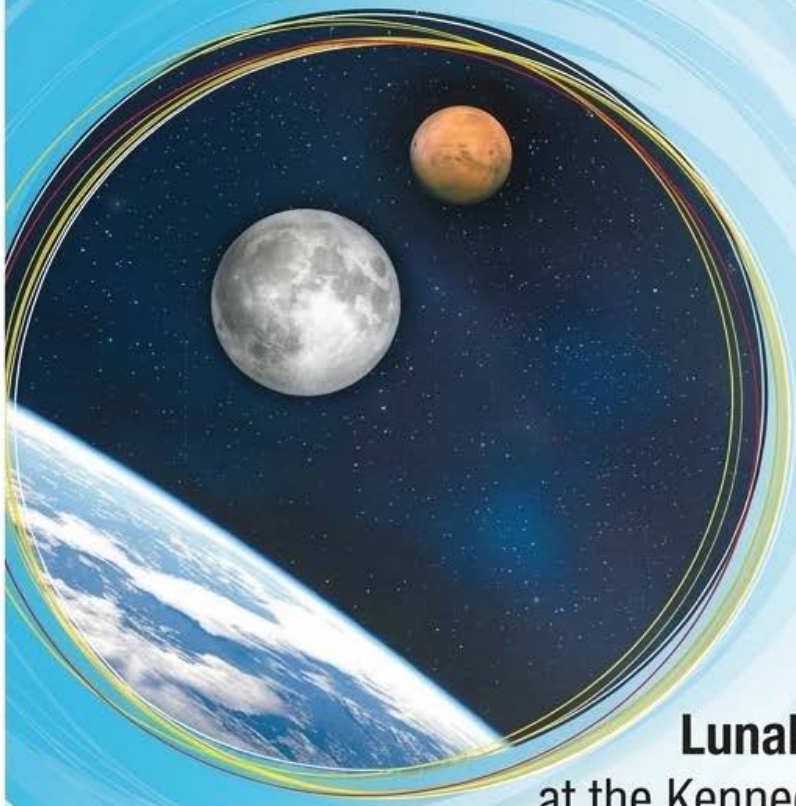
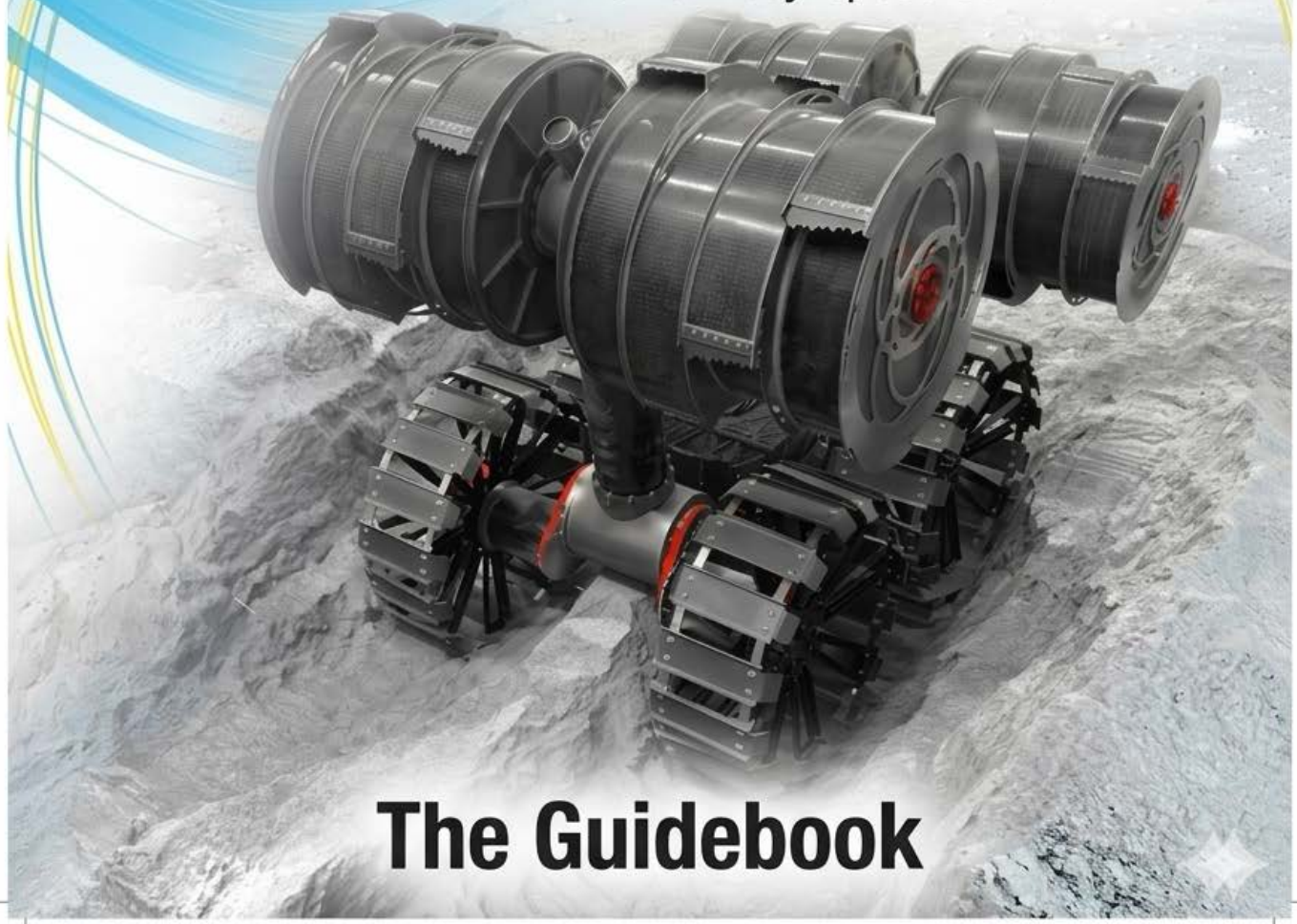


NASA's Lunabotics Challenge 2027

at the Kennedy Space Center



The Guidebook

CHANGE LOG

1. (10/5/26) UPDATED 3.4 Timeline with new dates.

All deliverables are due by noon Eastern Time on appointed date. NASA recognizes that there will be conflicts with other important school events and offers proctoring services if needed. Dates are subject to change.

October 13, 2026	Project Management Plan Due
October 13, 2026	Statement of Support – Institution
October 13, 2026	Statement of Support – Faculty
October 13, 2026	Statement of Rights of Use

2. (10/5/26) Updated and changed link to “Lunabotics Challenge website” to https://amfcse.org/lunabotics in all sections of the guidebook related to submissions

3. (10/5/26) Updated 4.2.2 – “Must be currently enrolled at the institution during the current semester or the one preceding the 2027 Lunabotics competition at Kennedy Space Center.”

4. (10/5/26) added – 4.1.6 - An institution may submit applications for multiple teams. Such teams must complete separate applications.

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3 NASA Lunabotics Challenge

3.1 Background

Since its inception in 2010, the NASA Lunabotics Challenge has become one of the agency's premier collegiate engineering competitions, culminating each year in an in-person event at NASA's Kennedy Space Center. Lunabotics directly supports the Moon Base Program within NASA's Human Spaceflight Mission Directorate (HSMD) by engaging student teams in the design, construction, and operation of prototype lunar robots capable of autonomously building berm structures using simulated lunar regolith.

These berms address several critical challenges associated with NASA's Artemis missions, including mitigating plume blast and ejecta during lunar landings and launches, supporting thermal management for cryogenic systems, providing radiation shielding for human habitats and sensitive infrastructure, and enabling other mission-critical surface operations. Through these hands-on systems engineering experiences, Lunabotics helps advance innovative lunar construction technologies while developing the next generation of engineers, scientists, and explorers.

NASA's Moon Base Program is focused on returning to the Moon to stay by establishing an enduring human presence through a phased, iterative approach that enables NASA to learn, adapt, and mature the technologies and operations required for future crewed missions to Mars. The program aims to develop a sustainable lunar outpost that advances scientific discovery, stimulates a growing lunar economy, and serves as a proving ground for deep space exploration systems.

3.2 Goals and Objectives

Lunabotics aims to advance the development of innovative, collaborative, and autonomous technologies through the NASA Lunabotics Challenge in support of NASA's Artemis missions and sustainable lunar exploration as a proving ground for future human missions to Mars. The challenge fosters partnerships among government, academia, and industry while addressing critical engineering challenges such as excavation, mobility, autonomy, and long-term system reliability in extreme environments. Lunabotics also helps cultivate a highly skilled 21st-century workforce by engaging students in a year-long, hands-on systems engineering experience that develops practical, technical, operational, and teamwork skills needed for successful careers in engineering and space exploration. Together, these efforts expand humanity's ability to explore the Moon, Mars, and beyond while delivering meaningful benefits on Earth.

3.3 Lunabotics Location

The competition will be held at the Astronauts Memorial Foundation's (AMF) Center for Space Education (CSE) adjacent to the Kennedy Space Center Visitor Complex (KSCVC).

Address: 405 Nasa Pkwy W, Merritt Island, FL 32953

3.4 Timeline

All deliverables are due by noon Eastern Time on appointed date. NASA recognizes that there will be conflicts with other important school events and offers proctoring services if needed. Dates are subject to change.

October 13, 2026	Project Management Plan Due
October 13, 2026	Statement of Support – Institution
October 13, 2026	Statement of Support – Faculty
October 13, 2026	Statement of Rights of Use
November 20, 2026	Preliminary Design Review Report Due
December 14, 2026	Team Selections Announced
January 15, 2027	Student and Faculty Registration Due
January 15, 2027	NASA Media Release Due
January 15, 2027	KSC Lunabotics Regulated Wast Management Overview Due
April 8, 2027	Systems Engineering Paper Due
April 29, 2027	Robot Data Report Due
April 29, 2027	Proof of Life Video Due
May 20-21, 2027	Practice Runs at KSC
May 24-27, 2027	Construction Runs at KSC

3.5 New for 2027

WE HEARD YOU!

NASA Lunabotics received significant feedback from participating universities that the 2026 Qualification and Finals structure required approximately two weeks of travel, competition participation, and associated logistics. Teams indicated that the length of the competition created challenges related to academic schedules, faculty availability, travel costs, internships, research obligations, and institutional travel approvals.

For 2027 the competition will return to a one-week competition at the Astronauts Memorial Foundation's (AMF) Center for Space Education (CSE) venue in the KSC Visitors Complex – there will be no Qualification activity at the University of Central Florida. All teams will compete at KSC.

3.6 Advisory

Kennedy Space Center is an active launch facility. As such, NASA mission priorities may necessitate changes to deadlines, deliverables, or event scheduling, including possible cancellation, without prior notice.

3.7 Glossary of Terms

1. **Autonomous:** The operation of a robot with no human interaction.
2. **Autonomy Attempt**: The period of time between when a team initiates their robot's autonomy mode and when they resume manual control of it.
3. **Autonomy Cycle**: The execution of a single autonomous operation (e.g. dig) from start to finish.
4. **Basaltic Regolith Properties:** Since the properties of Lunar regolith vary and are not well known, this competition will assume that basaltic regolith properties are similar to the Lunar regolith as stated in the "Lunar Sourcebook: A User's Guide to the Moon", edited by G. H. Heiken, D. T. Vaniman, and B. M. French, copyright 1991, Cambridge University Press (https://www.lpi.usra.edu/lunar_sourcebook/).
5. **CE EN166:** The core European technical standard that specifies the minimum safety and performance requirements for personal eye protection, including safety glasses, goggles, and face shields.
6. **I-551:** Form I-551 is the official government designation for a U.S. Permanent Resident Card, more commonly known as a green card.
7. **In-Situ Resource Utilization (ISRU)**: ISRU is the harnessing of local natural resources at mission destinations, instead of taking all needed supplies from Earth, to enhance the capabilities of human exploration. To live and work in deep space for months or years may mean crew members have less immediate access to the life-sustaining elements and critical supplies readily available on Earth. However, the farther humans go into deep space, the more important it will be to generate products with local materials. For more information, refer to https://beta.nasa.gov/mission/in-situ-resource-utilization-isru/.
8. **Kennedy Space Center Visitor Complex (KSCVC)** As a Smithsonian Affiliate, the KSCVC offers the chance to view artifacts of NASA's Mercury, Gemini, Apollo and Space Shuttle Programs in the context of exhibits and attractions that tell the NASA story. For more information, refer to https://www.kennedyspacecenter.com/.

9. **Lunar Regolith Density:** The density of regolith at the Apollo 15 landing site averages approximately 1.35 g/cm3 for the top 30 cm, and it is approximately 1.85g/cm3 at a depth of 60 cm. The regolith also includes breccia and rock fragments from the local bedrock. About half the weight of lunar soil is less than 60 to 80 microns in size.
10. **Mission Control Center (MCC):** Mission Control Center is the operations area where teams will teleoperate or autonomously control their robotic excavator to simulate a lunar In-Situ Resource Utilization (ISRU) construction mission. It is located outside of the Artemis Arena. The designated area used by a team's robot operator(s) to remotely execute their construction mission.
11. **Mission Cycle**: The sequence of robot operations conducted during a run: Excavate regolith -> Traverse obstacle zone loaded -> Deposit regolith - >Traverse obstacle zone empty -> Repeat.
12. **Regolith Construction Robot:** An autonomous or teleoperated robotic excavator including mechanical and electrical equipment, batteries, gases, fluids, and consumables delivered by a team to compete in the competition
13. **Regolith Construction Points:** Points earned during each construction run in the Artemis Arena. These points will be used to determine ranking in the on-site robotic operations category.
14. **Reference Arrow:** A clearly marked indicator on the mining robot, represented by an arrow denoting the robot's nominal forward direction. This arrow functions as a positional datum used to verify and validate the robot's initial orientation within the mining arena at the start of operations.
15. **RoboPits:** The area where teams will work on their robots, meet other competitors, and have their robot inspected before it goes to work.
16. **Run**: The period of time a robot is in the Artemis Arena to execute its berm building mission
17. **Telerobotic:** the area of robotics concerned with the control of semi-autonomous robots from a distance, chiefly using television, wireless networks (like Wi-Fi, Bluetooth and the Deep Space Network) or tethered connections.
18. Z87.1 refers to the ANSI/ISEA Z87.1 standard, the U.S. safety specification for occupational and educational eye and face protection, with the current edition being ANSI/ISEA Z87.1-2025.

3.8 Acronyms

ABET - Accreditation Board for Engineering and Technology

AMF - Astronauts Memorial Foundation

AS/NZS - Australian and New Zealand Standards

BC - Berm Construction

BCP – Berm Construction Productivity

BP-1 – Black Point 1

CAA – Central Accumulation Area

CDR – Critical Design Review

CFR – Code of Federal Regulations

COTS – Commercial off the Shelf

CSA – Canadian Standards Association

CSE – Center for Space Education

DR – Design Review

E-STOP – Emergency Stop

EIRP - Effective Isotropic Radiated Power

EMO – Emergency Off

EN166 – European Norm 166

FDEP - Florida Department of Environmental Protection

HSMD - Human Spaceflight Mission Directorate

IPA – Isopropyl Alcohol Wipes

ISM – Industrial, Scientific, and Medical

ISRU – In-Situ Resource Utilization

KSC – Kennedy Space Center

KSCVC – Kennedy Space Center Visitors Complex

MCC – Mission Control Center

MCD – Mission Control Director

MCJ – Mission Control Judge

NEMA - National Electrical Manufacturers Association

NFPA - National Fire Protection Agency

NiMH - Nickel Metal Hydride

OSHA – Occupational Safety and Health Administration

PB – Petroleum-Based

PDL - Prototype Development Laboratory

PDR – Preliminary Design Review

PIV - Personal Identity Verification

PMP – Project Management Plan

PPE – Personal Protective Equipment

PTB – Pre-Task Briefing

RCRA - Resource Conservation and Recovery Act

RTM - Requirements Traceability Matrix

RWMO - Regulated Waste Management Overview

SAA – Satellite Accumulation Area

SE – Systems Engineering

SEP – Systems Engineering Paper

SLA - Sealed Lead-Acid

SRR – Systems Requirements Review

SSID - Service Set Identifier

SWO – Stop Work Order

TPM - Technical Performance Measurements

UCF – University of Central Florida

UL - Underwriters Laboratories

WAP – Wireless Access Point

4 Eligibility and Participation Requirements

Lunabotics is open to students enrolled in post-secondary education or training in the United States, its Commonwealths, territories, or possessions. The following institutions are eligible to apply:

1. Community and State Colleges
2. Colleges and Universities
3. Career and technical education programs, and trade schools

4.1 Teams

1. Must be supported by the institution
2. Must be supported by a Faculty or School Advisor
3. Must include a minimum of two (2) undergraduate students
4. Must develop their own working robot
5. Two schools may collaborate to form one challenge team
6. An institution may submit applications for multiple teams. Such teams must complete separate applications.

4.2 Students

1. Must be 18 years old to attend the on-site events
2. Must be currently enrolled at the institution during the current semester or the one preceding the 2027 Lunabotics competition at Kennedy Space Center.
3. Can only be on one team

4.3 Faculty or Advisor

1. The advisor must be employed by the institution, authorized in writing or email, to represent the institution
2. Each team must be accompanied by an adult faculty/advisor age 21 or older
3. Must check-in with the team and stay with the team throughout the event(s)
4. Teams unable to meet this requirement will be disqualified from the challenge

4.4 Additional Requirements

1. All team members must be actively engaged.
2. Students must complete 100% of the project work. Professional assistance is permitted only for training or safety purposes

5 Application Process

The faculty advisor or team lead will initiate the application process by submitting the following deliverables at the AMF at Lunabotics Challenge website by Oct 13, 2026.

1. Project Management Plan (PMP) – See section 18.2 Rubric - Project Management Plan (PMP)
2. Statement of Support – Institution
3. Statement of Support – Faculty
4. Statement of Rights of Use

5.1 Institution Statement of Support

A statement of support from the institution, on institution letterhead indicating approval for the team and faculty advisor to participate in NASA's Lunabotics.

5.2 Faculty/Advisor Statement of Support

A statement of support from a faculty advisor on institution letterhead indicating a willingness to supervise and work with the team during all stages of NASA's Lunabotics. The faculty advisor must also sign off on the cover of all deliverables as evidence they have seen the deliverable and approves of the submission. There will be no consideration for teams working without a faculty advisor.

5.3 Statement of Rights of Use

The statement grants NASA, acting on behalf of the U.S. Government, rights to use the team's technical data and design concept, in part or in entirety, for government purposes. This statement is required. Teams will not be allowed to compete without this statement.

5.3.1 Statement of Rights of Use

As a team member for a(n) application entitled “NASA Lunabotics Challenge” proposed by a team of students the undersigned will and hereby do grant the U.S. Government a royalty-free, nonexclusive and irrevocable license to use, reproduce, distribute (including distribution by transmission) to the public, perform publicly, prepare derivative works, and display publicly, any data contained in this application in whole or in part and in any manner for Federal purposes and to have or permit others to do so for Federal purposes only.

In addition, the undersigned grants the U.S. Government a nonexclusive, nontransferable, irrevocable, paid-up worldwide license to practice, or have practiced on its behalf, any invention described in or arising from the application.

Name: _____

Title: _____

University or College: _____

Signature and Date: _____

List of Faculty & Student’s Names, Signatures and Dates

5.4 Selection Process

A Preliminary Design Review (PDR) down select process will be implemented for the 2027 competition. The Preliminary Design Review Report is due November 20, 2026. See 18.3 Rubric - Preliminary Design Review Report Rubric - Preliminary Design Review Report.

Applying does not guarantee acceptance into the Lunabotics Challenge. The team will be notified of the status via email. Selections will be announced December 14, 2026, per section 3.4 Timeline.

6 Requirements for Selected Teams

Teams that receive an email confirming their selection to the Lunabotics Challenge shall submit the following items and deliverables at AMF Lunabotics Challenge website to stay in the Challenge. Failure to submit any item in this section will result in removal from the challenge. NASA reserves the right to question or reject any item submitted by unauthorized representatives. Dates are subject to change. Deadline for all deliverables is **noon Eastern Time**. Deliverables are accepted up to seven (7) days prior to each deadline. Refer to the applicable sections for requirements and rubrics of each deliverable.

1. Student and Faculty Registration
2. NASA Media Release
3. KSC Lunabotics Regulated Waste Management Overview
4. Systems Engineering Paper
5. Robot Data Report
6. Proof of Life Video

6.1 Student and Faculty Registration

Student and faculty registration is required for all participants of the Lunabotics Challenge. Participants must register at the AMF Lunabotics Challenge website by January 15, 2027.

6.2 Media Release Authorization

All participants and visitors to Lunabotics events at Kennedy Space Center grant permission to be photographed and videotaped by NASA or its representatives for potential use in media products, unconditionally releasing NASA from related claims. Submit media release forms at the AMF Lunabotics Challenge website. Submission is required by January 15, 2027.

NASA Media Release for Adults

(Do Not Use for Minors)

I, _____ do hereby give permission to be
(Please print name your name)
interviewed, photographed, and/or videotaped by NASA or its representatives in connection with a NASA production.

I understand and agree that the text, photographs, and/or videotapes thereof containing my name, likeness, and voice, including transcripts thereof, may be used in the production of instructional, promotional materials, and for other purposes that NASA deems appropriate and that such materials may be distributed to the public and displayed publicly one or more times and in different formats, including but not limited to, websites, cablecasting, broadcasting, and other forms of transmission to the public. I also understand that this permission to use the text, photographs, videotapes, and name in such material is not limited in time and that I will not receive any compensation for granting this permission.

I understand that NASA has no obligation to use my name, likeness, or voice in the materials it produces, but if NASA so decides to use them, I acknowledge that it may edit such materials. I hereby waive the right to inspect or approve any such use, either in advance or following distribution or display.

I hereby unconditionally release NASA and its representatives from any and all claims and demands arising out of the activities authorized under the terms of this agreement.

By signing below, I represent that I am of legal age, have full legal capacity, and agree that I will not revoke or deny this agreement at any time.

I have read the foregoing and fully understand its contents.

Accepted by:

Signature: _____ Date: _____

Name and Location of Event: _____

Address: _____

Telephone: _____

Email Address: _____

6.3 Robot Data Report

All teams are required to provide technical information about their robot(s) via the Lunabotics Robot Data Form. This deliverable must be submitted at the AMF Lunabotics Challenge website. Teams with multiple robots must submit a separate form for each one. All form submissions are due by April 29, 2027.

6.4 Proof of Life Video

Prior to the on-site event, teams must provide video evidence to NASA judging staff that there is a safe and functioning robot to bring to KSC. The Proof of Life video must:

1. Show the faculty advisor verifying with a measuring tape that the robot meets the volumetric dimensions requirement: robot(s) shall be contained within a payload envelope measuring 150 cm length x 75 cm width x 75 cm height (vol. < 843,750 cm^3).)

   Length (cm) _____ X Width (cm) _____ X Height (cm)

2. Show the faculty advisor verifying with a weight scale that the robot meets the mass requirement (maximum mass of 80kg.)

   Robot Mass = _____(kg)

3. Show the location of all the data loggers on the robot(s).
4. Show the Commercial Off-The-Shelf (COTS) E-STOP red button with a minimum diameter of 4 cm.
5. Demonstrate the router(s) can turn on/off the 2.4 GHz band.
6. Include the following operations: traversing, excavating and dumping.
   a. The operations area can consist of regolith simulant, beach sand, play sand, construction sand, or outdoor volleyball sand as an acceptable granular material. If weather or other issues prevent operations on a granular surface, teams should use their best judgment to record operations.

Teams shall upload their videos to YouTube and submit a link to their recording at the AMF Lunabotics Challenge website. Teams who do not have a complete robot by the due date should present what is available.

Proof of Life submission is required by April 29.

6.5 Optional Deliverables

Team Photos may be uploaded at any time to https://www.facebook.com/LogansRun1976.

7 Event Safety and Awareness

7.1 Emergency Response Procedures

- Remain calm and report any hazardous or concerning situation.
- Notify the Pit Boss, Arena Chief, Mission Control Director, or Lunabotics Staff immediately.
- When uncertain, contact KSC First Responders at 321-867-7911. The facility location is the Astronauts Memorial Foundation's Center for Space Education, Building M6-306 (west side of the Kennedy Space Center Visitor Complex).
- Designate personnel to meet and guide First Responders to the incident location.

- Activation of First Responders is always acceptable; no judgment is associated with initiating a call.
- If instructed to evacuate the facility, all activity must cease and the facility must be exited promptly.
- The Marshaling Area is located adjacent to the Rocket Garden.
  Any incidents must be reported to KSC First Responders without delay.

7.2 BP-1 Safety Considerations

- BP-1 is alkaline and may cause skin and eye irritation
- BP-1 contains a small percentage of crystalline silica, which is a respiratory irritant
- If a person is allergic to talcum powder, they may be allergic to BP-1
- All participants must wear ANSI-approved, UL-Listed, CE EN166 rated, AS/NZS certified, or CSA rated PPE when in contact with BP-1
- Respiratory protection is required

7.3 Drones

Drones are strictly prohibited and will be confiscated if operated on Kennedy Space Center property.

If a team is found in violation of the Kennedy Space Center “NO DRONE POLICY”, Security will be notified and the team will be disqualified and removed from the venue immediately.

There is no exception to this regulation.

7.4 Prohibited Items

The Kennedy Space Center is a federal government facility. For the safety of visitors and employees, the following items are not permitted at Kennedy Space Center Visitor Complex:

- Firearms of any type, including those belonging to off-duty law enforcement (contact KSC Protective Service for more information)
- Ammunition of any type
- Pepper or Mace sprays
- Knives of any type, including any item with a blade
- Glass containers
- Alcoholic beverages and Controlled Substances
- Large coolers, luggage and other large bags

7.5 Fire Exits and Eyewash Stations

It is the responsibility of Lunabotics participants and guests to know where the fire exits, fire extinguishers, and eyewash stations are located.

7.6 Military Containers (ammo cans)

Spray paint or cover up the former military content signage to avoid any extra security checks.

7.7 Florida Weather

Lunabotics participants should come prepared for the Florida weather. Hats, sunglasses, insect repellent, sunscreen (SPF 50 or better), and a raincoat or poncho are recommended.

Teams are advised that their robots also will be exposed to the heat, humidity, wind, and rain. It is the sole responsibility of the team to protect their robot(s) from these elements.

Weather alerts for the following lightning conditions will be announced on the public address system:

- Phase I Lightning Condition - prepare to seek shelter.
- Phase II Lightning Condition - **seek shelter NOW** in any building.

7.8 Florida Wildlife

Alligators, ants, armadillos, mosquitoes, raccoons, snakes, wild hogs, etc. are common at KSC. Attempting to feed or interact with wildlife in any manner is prohibited.

8 At The Competition

A team found in violation of the rules, exhibiting unprofessional behavior or unsportsmanlike conduct, or not following the directions of the Lunabotics staff may be assessed penalty points, may be disqualified from a construction run, or disqualified from the entire Lunabotics Challenge.

8.1 Team Check-In

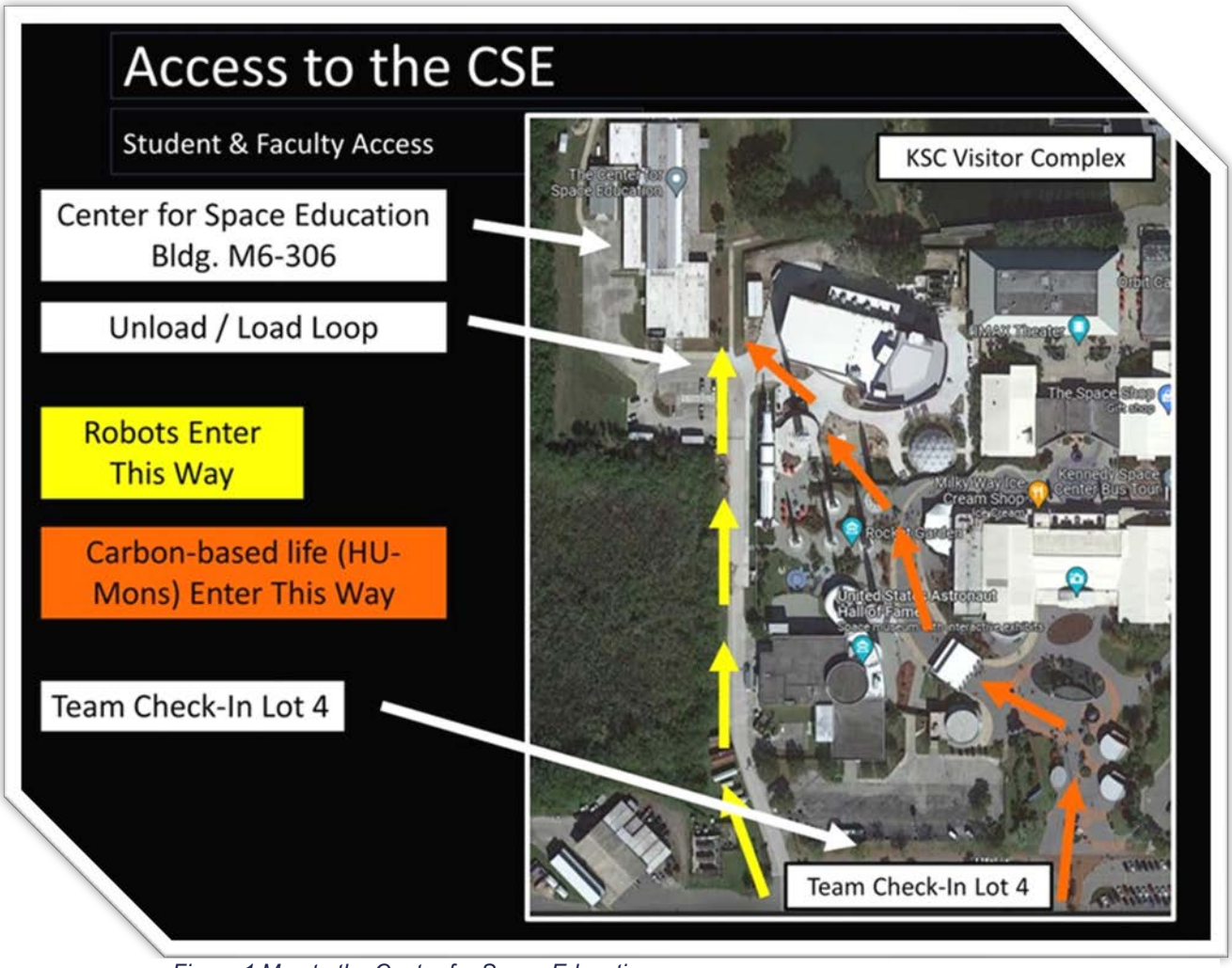


Figure 1 Map to the Center for Space Education

8.1.1 Initial Check-In 0700 to 0900

Parking passes will be sent via email to the faculty advisor and team lead of record. Drive through the KSCVC entrance, show the parking pass and proceed to Parking Lot 4.

1. At the check in table, the faculty advisor and team members shall check in together at the same time, with no exceptions.
2. If a name is not on the Event Check-In List, the team will be asked to stand by until the issue is resolved.
3. Check-in staff will provide a placard for vehicle and trailer to enter the parking lot of the AMF CSE. Students will leave robots, equipment, tools, nibblers, moopsies, etc., in the vehicle and trailer. The driver will remain with the vehicle and comply with the directions from security.
4. Check-in staff will direct the students to walk to the KSCVC Main Entrance, proceed through security, and then walk to the Center for Space Education Building (CSE) unloading area.
5. Students will meet their vehicle at the CSE unloading area, and proceed to the RoboPits to meet with the Pit Boss for further directions.

8.1.2 The following photo identifications are acceptable:

- Foreign passport.
- Permanent resident card or alien registration receipt card (Form I-551).
- Personal Identity Verification (PIV) card (smart card issued by the U.S. federal government)
- State-issued driver's license or ID card
- Tribal ID card issued by a federally recognized Native American tribe
- U.S. military dependent's ID card
- U.S. military ID card
- U.S. passport

8.1.3 The following photo identifications are not acceptable:

- School ID cards.
- Damaged ID cards, blurred images.

8.1.4 General Information

- The RoboPits open daily at 0700 and close at 1800.
- There will be various seminars and workshops throughout the week that may include systems engineering, autonomy, tech transfer, universities and NASA Internships.
- Meals and beverages are not provided by the event. Participants can purchase food items from the KSCVC and food trucks on-site.
- There is only one robot check-in. Robots can check out at any time but will not be allowed back into the facility.

8.2 Astronauts Memorial Foundation (AMF) Center for Space Education (CSE) Facility Layout

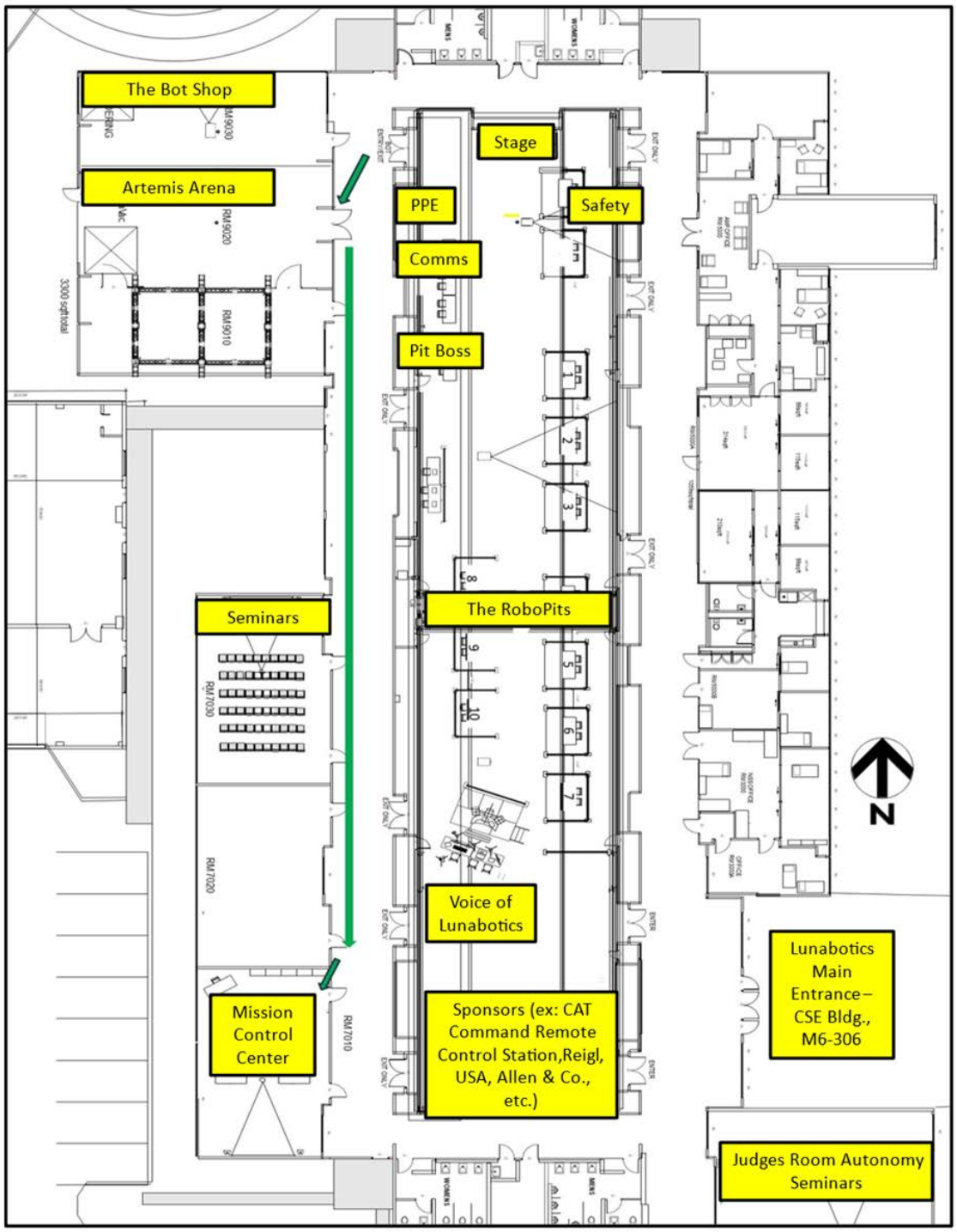


Figure 2 Astronauts Memorial Foundation Center for Space Education Facility Layout

Table 1 Astronauts Memorial Foundation (AMF) Center for Space Education (CSE) Room Assignments

Room No.	Operational Area
1005	Judge's Room
1000	Seminars
1001	Supplies & Staging
1005	Lunabotics Operations
2000	The RoboPits
7010	Mission Control Center
7030	Seminars
9010	Artemis Construction Arena
9020	Arena Operations
9030	The Bot Shop

9 The RoboPits

The RoboPits are the primary team workspace where competitors from across the country will prepare, test, and make final adjustments to their robots before entering the Artemis Arena. The RoboPits also serve as the main team gathering and collaboration area. For safety and traffic flow, teams and participants should not sit or congregate in the hallways outside the RoboPit area.

The RoboPit Room is equipped with emergency eyewash stations and designated disposal containers for used aerosol cans, batteries, degreasers, and cleaning wipes. Teams are encouraged to bring supplemental LED lighting, power strips, and a basic first-aid kit to support their operations throughout the competition.

Teams are responsible for creating their own pit identification signs. Be creative, use good judgment, and keep it fun. Signs must be no taller than 6 feet and may include LED light strings. Please note that Kennedy Space Center Fire and Safety personnel reserve the right to require modifications or removal of any sign or display determined to present a safety concern.

9.1 RoboPit Check-in and Setup

On day 1, teams must go to the Pit Boss Table to check in. The Pit Boss will assign a pit location to the team and explain the inspection process, robot run procedures, and other competition protocols. Please remain professional and patient during busy periods.

9.2 Contact Information

Teams must provide two contact phone numbers for emergency communication during the competition. These numbers will remain confidential and be deleted at the conclusion of the competition.

RobotPit Check-In Date: _____

School Name: _____

Faculty Name _____

*P.O.C. Cell No. _____

(* To contact the team on any issues and will be deleted at the end of the event).

Team Lead _____

*Team Lead Cell No. _____

(* To contact the team on any issues and will be deleted at the end of the event).

10 Pit Specifications, Guidelines & Shared Equipment:

Teams are assigned a 10' x 10' pit including two chairs, one table, and two power outlets with provided power strips. Daisy-chaining power strips is prohibited. Teams must keep all members and equipment within assigned pit boundaries, ensuring walkways and hallways remain clear and unobstructed. Vacuums are provided for shared use by all teams. Teams must return vacuums to the designated area after use and notify the Pit Boss of any safety concerns or vacuums requiring cleaning.

10.1 KSC Regulated Waste Management Overview (RWMO)

Students and faculty advisors attending the challenge on-site must read the following sections. Signed RWMO Acknowledgement forms can be submitted at the AMF Lunabotics Challenge website.

This information is a basic overview of the KSC waste management policy provided by NASA's Environmental Protection Specialist Environmental Assurance Branch (SI-E2). It is not formal or official Resource Conservation and Recovery Act (RCRA) training.

KSC Hazardous Waste Subject Matter Expert – Al Gibson

- Will be in the pit area throughout the event
- Visiting each team

For questions or concerns, please contact:

albert.e.gibson@nasa.gov, 321-861-0863

10.1.1 Regulated Waste Protocol

Teams must comply with Federal and Kennedy Space Center hazardous and controlled waste program requirements. Coordination with the Pit Boss is required before disposing of any of the following items (marked containers will be provided):

1. Batteries (Alkaline, Lithium, Ni-Cd)
2. Spray paint, spray foam, spray adhesives
3. Oily wipes, IPA solvent wipes, acetone wipes
4. Solder waste
5. PVC cement and primer (brushes, wipes, cans)
6. Lubricants (WD40, PB Blaster, 3-in-1 oil)
7. Adhesives (super glue, epoxy tubes)
8. Aerosol cans
9. Any other items as required by regulations

10.1.2 Background & Regulatory Requirements

1. Hazardous Waste Laws, Regulations, & Agencies
2. Federal Law: Resource Conservation & Recovery Act (RCRA)
3. Regulations: Code of Federal Regulations, Title 40, Parts 240- 299 United States Environmental Protection Agency (EPA)
4. Regulations: Florida Administrative Code Chapter 62-730 Florida Department of Environmental Protection (FDEP)
5. EPA delegated its authority to enforce RCRA regulations to FDEP
6. FDEP performs inspections, permitting, & enforcement at KSC
7. Hazardous Waste Types
8. Characteristic (flammable, corrosive, toxic, or reactive)
9. Listed (from a specific source)
10. Hazardous Waste Violations
11. In 2021, Florida’s hazardous waste program completed 34 formal enforcement actions (consent orders) for a total of $821,851.75 in civil penalties.
12. Criminal penalties – fines and/or jail time

13. KSC “Large Quantity Generator” of Hazardous Waste
14. Stricter management and reporting requirements and must comply with 40 CFR 262.15 and 40 CFR 262.17
15. Increased oversight by regulatory agencies
16. Hazardous Waste Cradle-to-Grave Liability
17. Listed EPA Waste
18. Hazardous waste sites are either Satellite Accumulation Area (SAA) or Central Accumulation Area (CAA)
19. Controlled Waste
20. Non-hazardous wastes not regulated by other laws i.e., used oil dry.
21. Universal Waste
22. Certain wastes that are exempt from being managed as hazardous waste as long as they are recycled, reused, or reclaimed in a certain manner.
23. If not managed as universal waste, it falls under the hazardous waste rules

10.1.3 Examples of Waste

1. Hazardous Waste Examples
2. PVC primer, cleaner and cement
3. Hypergolic scrubber liquor
4. Spent isopropyl alcohol & cleaning solvents
5. Solvent wipes
6. Paint thinners & unused paint
7. Acetone
8. Super glue
9. Examples of Controlled Waste
10. Used oil & oily wipes
11. Petroleum contaminated material
12. Examples of Universal Waste
13. Aerosol cans
14. Batteries

10.1.4 Waste Site and Locations

1. There will be Satellite Accumulation Activities (SAA) waste sites in the RoboPits and in the Bot Shop.
2. Place all waste in the appropriate container. If not sure, see the Pit Boss.

10.1.5 Waste Generators

Lunabotics waste generators training requirements are to:

1. Know hazardous waste has been generated (hazardous material = hazardous waste)
2. Know the waste must be managed in accordance with the RCRA regulations, i.e., ensure proper handling so that it does not go in the trash or poured/flushed into the sewer
3. If not sure about the waste let the Pit Boss know
4. If you see an issue report it to the Pit Boss.
5. Always follow proper safety protocol and use material as defined by the manufacture.
6. If you don't know the KSC way, ask – we are here to help you.



Figure 3 SAA Waste Example 1



Figure 4 SAA Waste Example 2

KSC Regulated Waste Management Overview (RWMO) Acknowledgement

Signature Sheet for Faculty Advisor and Students Attending. By signing this sheet, you affirm you have read this document and you will abide by KSC policy and procures for handling hazardous waste while attending Lunabotics.

Any questions or concerns about KSC Hazardous & Controlled Waste Management should be address to Al Gibson, albert.e.gibson@nasa.gov, 321-861-086.

Print Name, legibly	Signature	Date

10.2 Battery Protocol

1. Lithium-Ion and Nic-Cad batteries used in robots must be attended while charging. Chargers shall be unplugged overnight.
2. Battery containers must be designed for safely storing, charging, and transporting lithium-ion batteries, or approved equivalent.
3. Batteries must be stored in upright containers; batteries cannot be in contact with each other.
4. Batteries that have been dropped must be inspected for damage and replaced as needed.
5. Do not store batteries that are hot to the touch after charging.
6. If a battery continues to feel hot after charging, if possible, remove from the building and place outside and notify NASA Fire as a non-emergency issue at 321.861-5050.

10.3 Stop Work Order (SWO)

Not following standards or regulations, such as those set by the Occupational Safety and Health Administration, is one of the clearest examples of unsafe practice because those rules are built from real incident data and risk analysis. Lunabotics staff may issue an SWO for any suspected safety issue. Teams must immediately stop all activity. The Faculty Advisor must meet with the Pit Boss to resolve the issue. The SWO remains in effect until the Pit Boss makes a final ruling.



Figure 5 Stop Work Order Example 1



Figure 6 Stop Work Order Example 2

10.4 Clean-up and Check-out

The RoboPit area must be kept clean and organized at the end of each day, with no equipment or materials extending beyond the designated pit boundaries. All equipment must be powered down and unplugged before departing for the evening. Teams are responsible for maintaining the cleanliness of both their pit and the surrounding workspace and should use the provided vacuums as needed. Teams are encouraged to bring floor coverings or mats to help simplify cleanup efforts. Each team is expected to leave its RoboPit in the same condition in which it was received and must request a final inspection from the Pit Boss prior to final departure.

10.5 KSC Prototype Development Laboratory (PDL) Bot Shop

The Prototype Development Laboratory (PDL) Bot Shop is staffed by Kennedy Space Center technicians with expertise in the design, fabrication, and testing of prototypes and specialized test equipment. This mobile machine shop is equipped with grinding, sanding, mini-mill and mini-lathe, band saw, drill press and a variety of hand tools; however, welding capabilities are not available. Protective eyewear must be worn in the Bot Shop at all times. The Bot Shop is intended to support teams with robot repairs and limited modifications and may not be used to complete unfinished robots.

The Bot Shop is busy throughout the competition. For more information, refer to https://public.ksc.nasa.gov/partnerships/capabilities-and-testing/testing-and-labs/prototypedevelopment-laboratory/.

10.6 Robot Inspections - Safety and Communications

1. Inspection stations will be clearly identified

2. Teams are encouraged to have initial safety and comm inspections as soon as possible.
3. Robots will have an initial safety inspection and an official safety inspection prior to each run, for up to a total of 4 inspections.
4. Initial inspections operate on a first-come, first-served basis (no scheduling)
5. For competition and practice runs, an escort will retrieve teams from their pit before each construction run for the official safety inspection—do not go to the inspection station without the escort.

ROBOTS MUST PASS BOTH SAFETY AND COMMUNICATIONS INSPECTIONS BEFORE ENTERING THE ARTEMIS ARENA.

10.7 Personal Protective Equipment (PPE) Robo Pits and Bot Shop

10.7.1 Eye and Face Protection

ANSI-approved protective eyewear shall be worn without exception in individual pits and the Bot Shop. Non-safety-rated prescription glasses require ANSI-approved safety goggles worn over them. Safety-rated prescription glasses marked with standards (e.g., Z87.1) may be used with appropriate side shields and frames.

NASA will provide standard safety glasses. Teams and schools will supply additional protection options including side shields and face shields. Additional protection is recommended for robot work including grinding, drilling, soldering, cutting, and activities with risk of flying particles or chemical exposure (splashes, splatters, sprays).

Reflective lenses are prohibited; eyes must be clearly visible.

10.7.2 Hair Requirements

Hair must not impede vision or contact work surfaces. Long hair must be tied back.

10.7.3 Hand Protection

Use appropriate gloves and mechanical tool guards to protect against heat, electrical, chemical, laceration, and mechanical hazards. Select proper protection for each specific activity.

10.7.4 Hearing Protection

Provide and use hearing protection devices (such as earplugs) in areas with objectionable or questionable sound levels.

10.7.5 Foot Protection

Wear completely enclosed shoes that cover the instep, preferably made of materials such as leather that can be wiped clean of BP-1 dust. Open-toe, open-back, open-weave shoes, or shoes with holes that could expose skin to BP-1 are prohibited.

10.7.6 Clothing Requirements

Clothing requirements are mandatory and include shirts or tops that fully cover the upper torso, floor-length pants that extend to the ankles, and baseball caps or other headgear positioned so as to not impair vision or interfere with protective eyewear. All participants must comply with these requirements at all times while on site.

10.7.7 Prohibited Clothing

The following items are prohibited while on site:

1. Flowing garments and loose neckwear (ties, scarves)
2. Caps worn low over eyes that impede vision
3. Cropped shirts, plunging necklines, spaghetti straps, or damaged shirts
4. Shorts, capris, skirts or damaged pants
5. Loose or wide, bell-sleeved tops, coats, or shawls

11 The Artemis Arena

11.1 Arena Composition

1. The arena contains Black Point-1 (BP-1), a crushed basalt lunar regolith simulant to a depth of approximately 45 cm. Coefficient of friction and cohesion are highly dependent on the humidity, bulk density and porosity. For more information, refer to https://ares.jsc.nasa.gov/projects/simulants/bp-1.html). Note the following:
    a. It does not behave like sand.
    b. The coefficient of friction has not been measured.
    c. There are naturally occurring rocks in the aggregate.
    d. BP-1 is made from crushed basalt fines and not commercially available.
    e. Reference Soil Test Apparatus for Lunar Surfaces.
    f. The density of the compacted BP-1 aggregate will be between 1.5 g/cm3 and 1.8 g/cm3.
    g. BP-1 behaves like a silty powder soil with most particles under 100 microns in diameter.
    h. Will be compacted and the top layer will be raked to a fluffy condition of approximately 0.75 g/cm3, similar to the Lunar surface.
    i. Teams are encouraged to develop or procure simulants based on basaltic minerals and lunar surface regolith particle size, shape, and distribution.
2. There is naturally occurring additional material randomly distributed that may be mixed with the BP-1. This gravel material may be approximately 2 cm diameter.
3. **BP-1 Reflectivity:** NASA performed tests to answer questions about BP-1 reflectivity for LIDAR (or other LASER-based) navigation systems. The laser is not a beam – it is spread out as a sheet that is oriented in the vertical direction, so it is draped across the BP-1 and across a white/gray/black target that is standing up behind the BP-1 in the images. The BP-1 is the mound at the bottom of each image. Teams can get the reflectivity of the BP-1 by comparing the brightness of the laser sheet seen reflected from the BP-1 with the brightness of the same sheet reflected from the white and black portions of the target.
    a. Note the BP-1 is mounded. Teams need to account for the fact that it is not a flat surface for analyzing the brightness in the images. The three pictures below were shot with the camera at 10, 16, and 21 degrees relative to the surface.

b. The laser was at a phase angle of 15 degrees. The camera speed and aperture were set to (manual mode): 1/8 s, f/4.5.

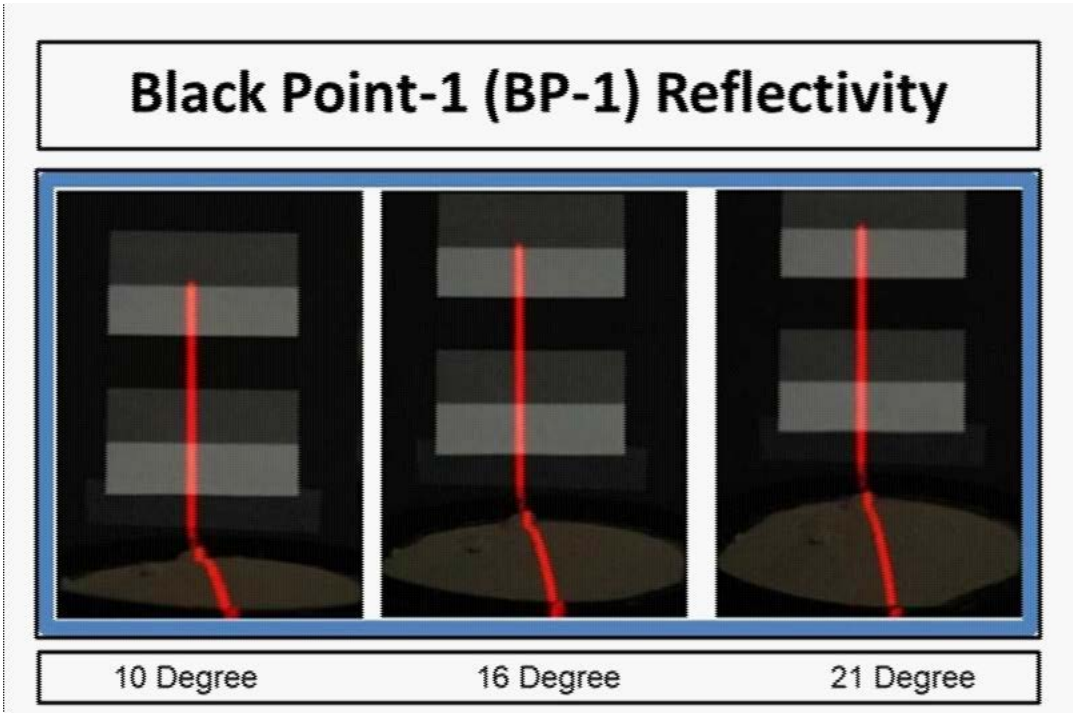


Figure 7 LIDAR/Laser Reflectivity

11.2 Arena Dimensions

1. The interior dimensions are approximately 6.8 m long × 5.0 m wide measured between the interior ducts. Reference Figure 9 Arena Dimensions and Obstacle Layout for the 17 cm diameter PVC pipe running along the inside perimeter.
2. The berm shall be built in a designated/marked area measuring 2.2m x 0.9m. Only berm volume in the red designated/marked box will be counted for scoring purposes. Reference Figure 8 Berm Positioning. Note: Only the green berm volume inside the red box will count towards the berm volume measurement.

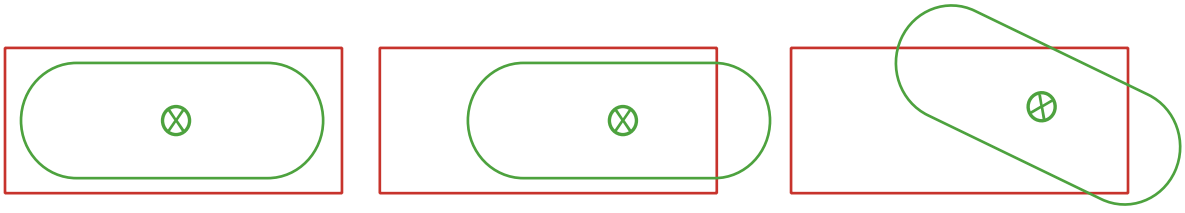


Figure 8 Berm Positioning

3. The berm placement and center location are referenced to the **arena wall**, consistent with the scoring definitions. The PVC ducts and any wall-mounted hardware, including 80/20 aluminum in the Starting Zone, are treated as physical arena infrastructure and not as geometric references for

berm placement, localization, or scoring. Center of Berm location X = 5.38m, Y = 0.9m from the Arena Wall in the corner of the starting zone.

11.3 Obstacle Zone

The obstacle zone is designed to test:

- Obstacle detection capabilities
- Mapping functions
- Navigation planning abilities

11.4 Obstacle Types

11.4.1 Boulder Obstacles

- Quantity: Minimum of three (3)
- Placement: Randomly positioned before each competition round
- Size: Approximately 30-40 cm in diameter with varying heights
- Note: Boulders may also appear in the excavation zone but will not exceed the dimensions of those in the obstacle zone.

11.4.2 Crater Obstacles

- Quantity: Minimum of three (3)
- Dimensions: Varying depths and widths, not exceeding ~ 40-50 cm in either measurement
- Permanent obstacle that must be avoided - central support column

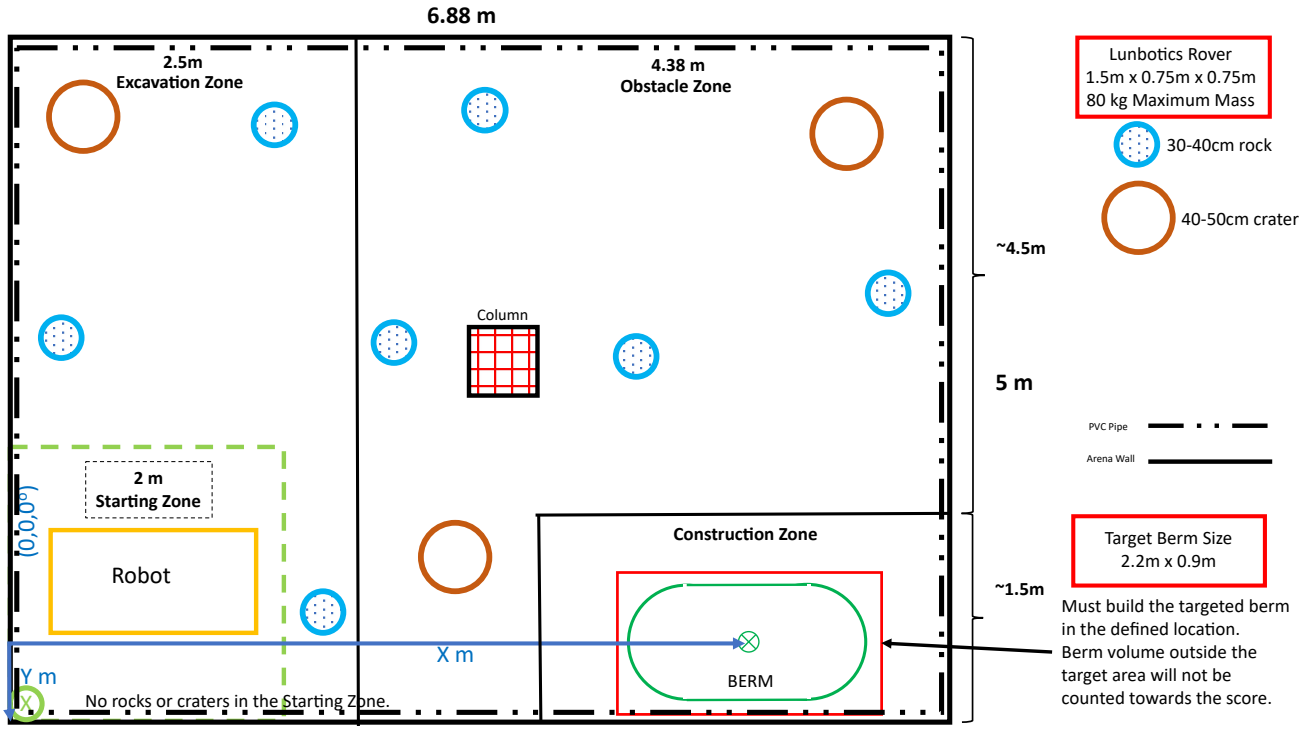


Figure 9 Arena Dimensions and Obstacle Layout

11.5 Artemis Arena Camera Layout

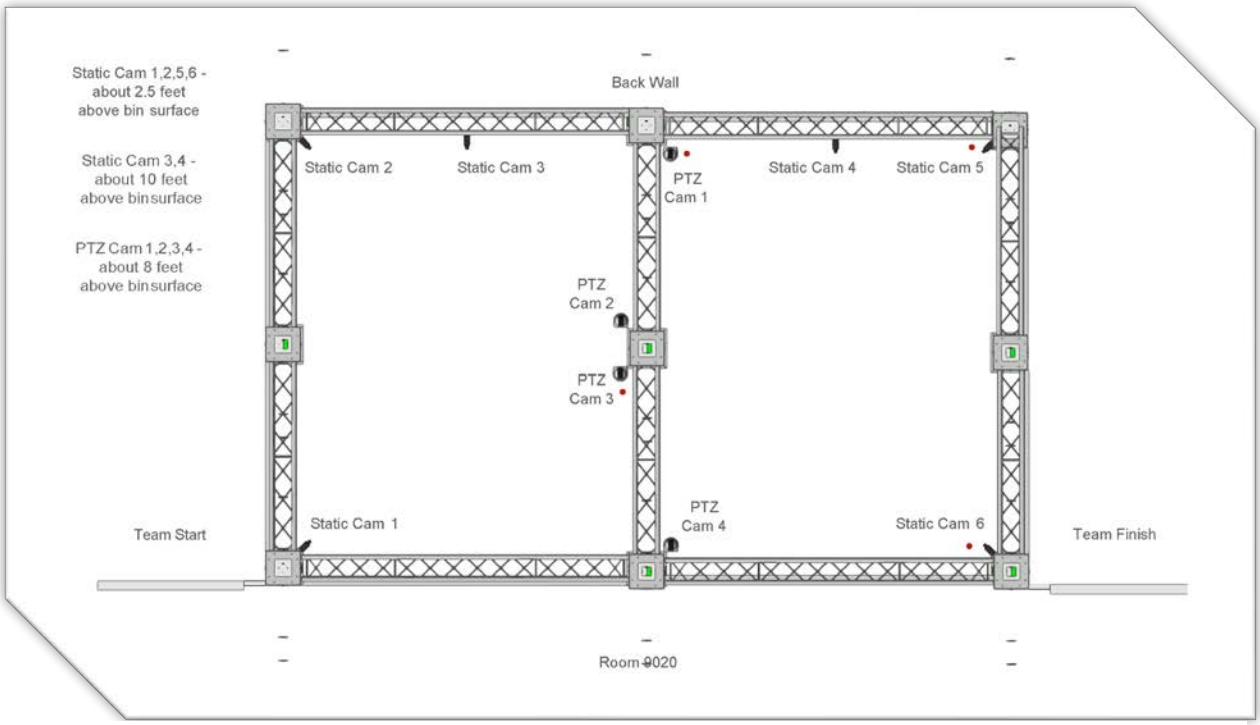


Figure 10 Arena Camera Layout

11.6 Artemis Arena Protocol and Requirements

1. The Artemis Arena Staging Area, Competition Arena, HEPA station, and Arena Chief Station are located within the Artemis Arena.
2. Students shall promptly follow the instructions of the Arena Chief and Arena personnel.
3. Faculty or advisors are not permitted in the Arena during the competition.
4. The Artemis Arena is considered an operational area with restricted access. Access is restricted to currently active competing team members only. If the team's robot is not in the arena or arena staging area, team members are not permitted in the Artemis Arena.
    a. For Arena operations, the currently active competing team members are defined as the team members attired in PPE that are tasked with placing the robot and router in the arena during the setup period and removing the robot and router from the arena after the run has ended. A maximum of 4 team members are permitted in the arena.
5. Photography and electronic devices (smart devices, tablets, cell phones, Meta Glasses, etc.) not required for robot operations are prohibited in the Artemis Arena. Violation of this rule may result in penalty points assessed or team disqualification.
6. Arena team members are prohibited from informing or pointing out obstacles, craters, and other arena conditions to the MCC team members.

7. Team members placing robots in the Competition Arena shall don PPE in the RoboPits per section 11.7.3 Donning Protocol.
8. The Arena Chief will conduct an overview Pre-Task Briefing (PTB) with the designated team members at the Artemis Arena entrance.
9. The Arena Chief will grant access to the Artemis Arena staging area when ready.
10. In the Artemis Arena staging area, the PPE attired team members will be given a Pre-Task Briefing (PTB) containing specific information needed for the run.
11. Team members in PPE shall remain with their robot in the designated Arena staging area until directed otherwise by the Arena Chief or designated representative.
12. Approaching the Competition Arena before being instructed to do so by the Arena Chief or designated representative is not permitted. Violation of this rule may result in penalty points assessed or team disqualification.
13. The Arena Chief will lead the team members through the Competition Arena door when the Competition Arena is ready for the team.
14. The team will point out the location and number of power meter or data logging devices on the robot to the arena judges.
15. At the direction of the Arena personnel, the team members will place their robot in the Arena and perform setup activities necessary to establish communication with Mission Control. The construction robot will be placed in the arena in the selected starting position and direction.
16. Only team members in PPE may observe the construction run from a designated area in the Arena. The team members must remain in the designated area for the duration of the run.
17. During the construction run, team members are not permitted to give hand signals or any other motions that could direct the team members in the MCC. Violation of this rule may result in penalty points assessed or team disqualification.
18. When the construction run has ended and when directed by arena personnel, the team members in PPE shall promptly retrieve their robot and equipment from the Arena and proceed to the HEPA station.
19. At the HEPA station, team members will vacuum excess BP-1 from their robot.
20. After vacuuming, team members will doff PPE in the Artemis Arena room
    a. Return goggles for cleaning and reuse.
    b. Remove hood then remove Tyvek coverall rolling down from inside of suit.
    c. Place coveralls and gloves in designated waste containers in the HEPA area. N-95 masks will remain on until outside of the Arena.
21. Remain in the HEPA area until directed by the Arena Chief or designated personnel.
22. When directed by Arena personnel, the team members will promptly exit the Arena. Place N-95 masks in the designated waste container at the Arena entrance on departure.

23. Remember, the Artemis Arena is an operational area during competition days. There are many activities occurring in series and in parallel. It is very important that everyone in the Arena practice situational awareness at all times.



Figure 11 Artemis Arena Judges

11.7 Personal Protective Equipment (PPE) Artemis Arena

The following PPE is required when entering the Artemis Arena. NASA will provide the standard PPE. Teams are responsible for providing additional PPE required by OSHA, ANSI, or equivalent standards to accommodate participants in this event.

1. Protective cover-all suits
2. Protective gloves
3. Protective booties
4. N-95 masks
5. Safety goggles

11.7.1 Respiratory Protection Requirements

All participants will follow OSHA 1910.134 - Occupational Safety and Health Standards, Subpart I: Personal Protective Equipment, Standard Number: 1910.134, Title: Respiratory Protection. Under OSHA 1910.134(a)(2): A respirator shall be provided to each employee when such equipment is necessary to protect the health of such employee.

11.7.2 Facial Hair Restrictions

Per OSHA Standard 1910.134, respirators cannot be worn when facial hair interferes with the sealing surface of the facepiece or valve function. While OSHA 1910.134 does not provide religious exemptions, loose-fitting hoods or helmets that accommodate facial hair are acceptable alternatives. Teams are responsible for providing specialized respiratory equipment required by regulation. Students that don't meet the OSHA regulation are not permitted into the arena.

11.7.3 Donning Protocol

1. Select and don appropriately sized protective coverall suits
2. Don appropriately sized gloves, overlapping the coverall suit
3. Tape gloves to coverall overlap to prevent skin exposure
4. Apply tape above ankles and waist as needed to secure excess material
5. Don protective booties
6. Place N-95 mask over nose and mouth, adjusting nose bridge for comfort and to prevent fogging
7. Place dust goggles over eyes

12 Official Competition Flow

Each team will have one 15 minute practice run and two, non-consecutive, 15 minute construction runs. The practice run will only be allowed after the robot has passed the initial Communications and Safety Inspections. The Safety Inspection will be repeated prior to each practice and construction run. When both inspections have passed the robot is ready for runs in the Artemis Arena.

For each practice and construction run, a Pit Crew member will get the team with their robot from their Pit approximately 45 minutes before their scheduled start time. At this point the teams and the robot will be escorted to the inspection station for the official safety inspection. At the same time, up to four (4) students will proceed to the PPE preparation area to don the required personnel protective equipment for the Artemis Arena.

An escort will then accompany the team to the Artemis Arena entrance for the Arena Chief briefing. **Important: Teams that are not ready or cannot be located at the designated time will forfeit their competition opportunity, and their slot will be given to another team.**

12.1 Robot Safety Inspections

Judge's Name: _____

School Name: _____

Practice Run ___ Competition Run 1 _____ Competition Run 2 _____ Date: _____

1. Identify All Data Loggers (Yes / No) _____
2. E-STOP Button/Kill Switch (Yes / No) _____
3. Lifting Points – Minimum or 4 (Yes / No) _____
4. Volumetric Dimensions - robot(s) shall be contained within a payload envelope measuring 150 cm length x 75 cm width x 75 cm height (vol. < 843,750 cm^3).

   Length _____ x Width _____ x Height

5. Robot Mass (maximum mass of 80kg. Robot Mass = _____
6. Dust Tolerant Design
   a. Drive train, components enclosed/protected, up to 15 points _____
   b. Active dust control - brushing, electrostatics, etc., up to 5 points _____
   c. Custom dust sealing features - bellows, seals, etc., up to 10 points _____

   Total Points (a+b+c) _____

The launch volume dimensions of the robot may be oriented in any way (length, width, height can be defined along any of the X, Y and Z-axis. The team must declare the robot orientation to the inspection judge by: Arrow 1 – length. Arrow 2 – width. Arrow 3 – marking the forward direction of the robot in the starting position. The reference location and arrow pointing forward can point any direction of the team's choosing, except up or down. The arrow is used only to orientate the robot prior to starting the robot run to face the robot arrow either north, east, south, or west after spinning the direction wheel.

==========

Rule Violation or Penalty Points _____

_____

_____

_____

_____

13 Robot Design and Operation Requirements

Components (i.e. electronic, mechanical, etc.) are not required to be space qualified for atmospheric, electromagnetic, thermal, or off-world environments. Bulk regolith construction requires teams to consider several design and operation factors such as robot dust tolerance, minimizing dust projection, efficient communications, minimal vehicle mass, minimal energy or power required and maximum autonomy.

If any inconsistencies arise between section 13 Robot Design and Operation Requirements and other sections, 13 Robot Design and Operation Requirements shall prevail.

13.1 Power

1. Each robot shall include a removable Commercial Off-The-Shelf (COTS) electronic data logger or device to record and display the energy consumed during each run. The devices should be placed where judges can easily reach and remove them without having to kneel or get down on the floor.

2. The data logger for power consumption shall be schematically located between the battery and kill switch, so the readings are not erased if the E-STOP button is activated.

3. A 30-point reduction to ***Berm Construction Productivity – Normalized for Energy (BCP Energy)*** will be accessed if the robot is not wired in accordance with this requirement. This will be checked at Inspection. Also, if a robot is not wired in accordance, teams run the risk of scoring zero if the E-STOP has to be triggered.

4. Actual energy consumed during each run shall be shown to the judges on the data logger immediately after the run. “Immediate” includes time for the judge to climb into the arena, find the logger, and record the power reading. If the logger is independently powered, then the robot can be remotely powered off after the run. Although this is acceptable, it is not recommended in case the robot needs to be commanded to complete an operation so that it can be removed from the arena.

5. The robot shall provide its own onboard power. No facility power will be provided to the robot during the run. There are no power limitations except that the robot must be self-powered and included in the maximum mass limit.

13.2 E-STOP Button / Kill Switch

1. Also known more formally as an emergency brake, emergency stop (E-stop), emergency off (EMO), kill switch, or emergency power off (EPO), is a safety mechanism used to shut off machinery in an emergency, when it cannot be shut down in the usual manner.

2. The robot shall be equipped with an unmodified Commercial Off-The-Shelf (COTS) E-STOP with a red button.

3. The E-STOP button shall have a minimum diameter of 40 mm; be located on the surface of the construction robot and require no additional steps to access it (OSHA, IEC 60204-1).

4. The E-STOP button shall be placed on the highest location on the robot.

5. The E-STOP button shall be accessible from any side of the robot.

6. The E-STOP button shall be unobstructed—no collars or actuation restrictions—and easily accessible without having to reach over, under or around to actuate. Machine-building such as

ANSI B11, B11-19 and National Fire Protection Agency (NFPA) 79 also address specifics in regard to safety devices such as an e-stop.

7. The E-STOP shall not automatically reset. OSHA and relevant standards such as IEC 60204-1 further state that resetting of the E-STOP alone shall not resume operation. A second deliberate action is needed, such as the pressing of a reset button. This could include twisting the mushroom button and allowing it to spring back up or pulling the button back up to reset.
8. The E-STOP button shall stop the construction robot's motion and disable power with one push motion on the button. It shall be reliable and instantaneous. A closed control signal to a mechanical relay is allowed as long as it stays open to disable the robot. This rule exists in order to have the capability to safe the construction robot in the event of a fire or other mishap. The button shall disconnect the batteries from all controllers (high current, forklift type button) and it shall isolate the batteries from the rest of the active sub-systems as well.
9. There shall be only one E-STOP button per robot and in the case of multiple robots, each robot shall have its own E-STOP button.
10. Disabling the E-STOP button without authorization from the Staff shall result in a safety disqualification.
11. Only onboard laptop computers may stay powered on if powered by their own, independent, internal computer battery and whose power does not flow through the main robot E-STOP button. For example: it is acceptable to have a small battery onboard that only powers a Raspberry Pi control computer, and whose power does not flow through the main robot E-STOP button.

13.3 Volumetric Dimensions

1. Robot(s) shall be contained within a payload envelope measuring 150 cm x 75 cm x 75 cm (vol. ≤ 843,750 cm3). Robots will be inspected for the volumetric dimensions in the stowed configuration during the safety Inspection. No modifications or team robot interaction is permitted during this verification.
2. The robot may deploy or expand beyond the initial envelope after the start of each run but shall not exceed 250 cm above the surface of the regolith.
3. Multiple Robot Systems are allowed but the total mass of the whole system shall comply with the mass given in this rule. The starting dimensions of the set of multiple robots shall comply with the volumetric dimensions measuring 1.50 m x 0.75 m x 0.75 m. This limit applies to the entire system, not to each individual robot. The robot can separate itself intentionally, if desired, but all parts of the robot must be under the team's control at all times. Unintentional breakage will not be counted against the team. The robot does not have to re-assemble prior to the end of the construction run.

13.4 Robot Mass

1. Robot mass cannot exceed the maximum mass of 80kg. A lower robot mass is beneficial to the construction score.
2. Robots shall have a minimum of four (4) lifting points (OSHA 1926), be safe for human hands and clearly marked for students and staff to use. Teams are responsible for placement and removal of their construction robot onto the BP-1 surface. There must be one person per 20 kg of mass of the

construction robot, requiring a minimum of four people to carry the maximum allowed mass of 80 kg.

3. Subsystems and equipment on the robot that are used to transmit commands and data and video to the telerobotic operators are counted toward the mass limit.
4. The mass of the navigational aid system, including any beacons or targets not attached to the robot, is included in the maximum construction robot mass limit. These items must be self-powered.
5. Equipment not on the robot used to receive data from and send commands to the robot for telerobotic operations is excluded from the mass limit.

13.5 Robot Orientation

The team must declare the robot orientation with three arrows. Arrow 1 declares the length. Arrow 2 declares the width. Arrow 3 must mark the forward direction of the mining robot in the starting position configuration. The reference location and arrow pointing forward can point any direction of the team's choosing, except up or down. Arrows on any location on the robot is acceptable if multiple arrows do not conflict. The arrow is used only to orientate the robot prior to starting the robot run to face the robot arrow either north, east, south, or west after spinning the direction wheel.

13.6 Honeycomb Structures

Teams may use honeycomb structures provided the design is strong enough to be safe and the edges are sealed to prevent dust intrusion. A wheel with a large honeycomb structure that is open on both sides is allowed if the edges are not sharp enough to create a cutting hazard.

13.7 Navigation Requirements

1. Global Positioning Satellite (GPS), IMU-enabled GPS devices and compasses are not allowed to contribute to the robot internal calculations. Teams must explain to the safety inspection judges how these features will be switched off, or how the data is subtracted to ensure the internal calculations do not make use of them.
2. Beacons or fiducial targets may be attached to the designated arena frame area for navigation purposes only. The designated area is anywhere on the bin frame structure along the perimeter of the starting zone (2 sides). Tape, clamps, or rods pushed into the regolith may be used, but screws or other fasteners requiring holes may not be used. This navigational aid system must be attached during the setup time and removed afterwards during the removal time during each run.
3. The target and beacon may be a passive fiducial. It may also send a signal or light beam or use a laser-based detection system which has not been modified (optics or power). Only Class I or Class II lasers, or low-powered lasers (< 5mW) are allowed. Supporting documentation from the laser instrumentation vendor must be provided to the inspection judges for "eye-safe" lasers.

13.8 Fundamental Physical Processes

1. Any fundamental physical process that would not work in an off-world environment will not pass inspection. Teams may not use GPS, rubber pneumatic tires, air or foam filled tires, open or closed cell foam, ultrasonic proximity sensors, or hydraulics because NASA does not anticipate the use of these on an off-world mission. For example, any dust removal from a lens or sensor must employ a physical process that would be suitable for the Lunar surface.

2. Closed loop pneumatic systems are permitted if the gas is supplied by the robot and self-contained. The rules are intended for robots to show an off-world plausible system functionality, but the components do not have to be traceable to an off-world qualified component version.

3. Teams may use processes that require an Earth-like environment (e.g., oxygen, water) only if the system using the processes is designed to work in a Lunar environment and if such resources used by the robot are included in the mass of the robot.

4. Examples of allowable components are: Sealed Lead-Acid (SLA) or Nickel Metal Hydride (NiMH) batteries, composite materials, rubber or plastic parts, actively fan cooled electronics, motors with brushes, infrared sensors, inertial measurement units, and proximity detectors and Hall Effect sensors, but proceed at your own risk since BP-1 regolith simulant are very dusty and abrasive.

13.9 Robotic Operation Requirements

1. Teams will be permitted to repair or otherwise modify their robots while the RoboPits are open. The robot will be inspected for safety before each competition run. Robots cannot be modified after their practice and competition run safety inspection.

2. Each team is allowed **up to** 10 minutes to place the construction robot in its designated starting position within the arena and perform required setups from MCC. The team is allowed 5 minutes to remove the robot after the run has ended as directed by the Arena Judges.

3. The robot's starting direction and location will be randomly selected immediately before the competition run.

4. The robot **shall not**:

    a. be anchored to the arena surface or regolith simulant prior to the beginning of each competition run.

    b. ram the wall, which may result in a safety disqualification for that run

    c. use any aspect of the arena (wall, structure, column, etc.) in attempting any operations.

5. At the start of the construction run, the mining robot shall not occupy any location outside the defined starting position in the Artemis Arena.

6. The robot shall operate within the Artemis Arena. It is not permitted to pass beyond the confines of the outside perimeter of the arena or hit the walls during the construction run.

7. The robot shall not use any process that causes the physical or chemical properties of the regolith simulant to be changed or otherwise endangers the uniformity between competition runs.

8. The mining robot shall not penetrate the regolith simulant surface with more force than the weight of the mining robot.

9. No ordnance, projectile, far-reaching mechanism, etc. shall be used. The mining robot must move on the regolith simulant surface. Far-reaching mechanism in this context does not include any deployed or extended component as allowed in the dimensions rule above. Those will not violate this rule.

10. The robot can run either telerobotically or autonomously.

11. The robot can excavate in the starting zone.
12. The robot may start construction operations as soon as any part of it crosses into the construction zone.
13. The robot shall move from the starting zone, into the excavation zone, and across the obstacle zone to the construction zone.
14. The robot shall not acquire regolith simulant for the berm from inside the obstacle zone or construction zone.
15. The robot shall not push or move any obstacles in the obstacle zone. The obstacles may only be pushed to the side of the arena in the construction and excavation zones.
16. The robot shall avoid the craters in the obstacle zone. It shall not fill in any craters.
17. Obstacles may not be moved into the excavation zone.
18. Bulldozing such as pushing a pile of dirt or rocks with a bladed dozer-type of rover is not considered an acceptable excavation and regolith simulant transfer technique for the Lunabotics Challenge. All regolith simulants must be carried through the obstacle zone.
19. The robot shall end operations immediately when the power-off command is sent or as instructed by the Arena Judge. After the official construction run ends, the arena judge will determine if the robot needs to move prior to being removed. After ensuring that the lidar scan of the berm has been completed, the judge will instruct the team members when they can enter the arena to remove the robot.

14 Communications Requirements

This configuration shall be used for teams to communicate with their robot. If any inconsistencies arise between section 14 Communications Requirements and other sections, 14 Communications Requirements shall prevail.

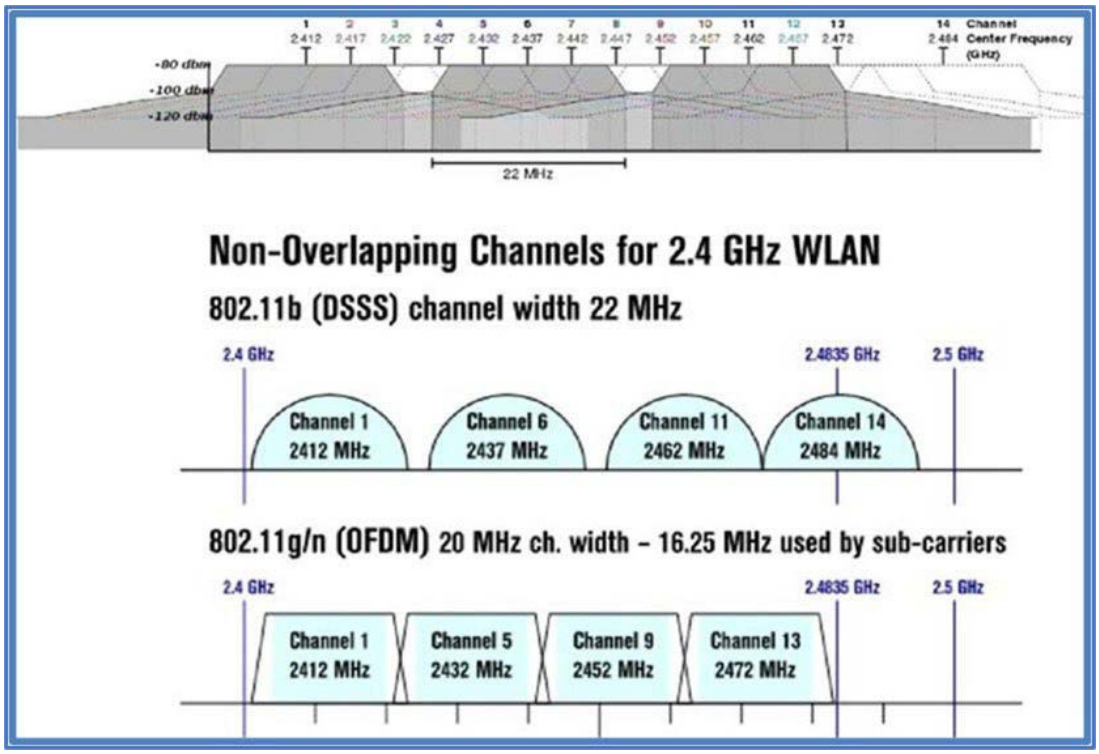


Figure 12 Communication Channels

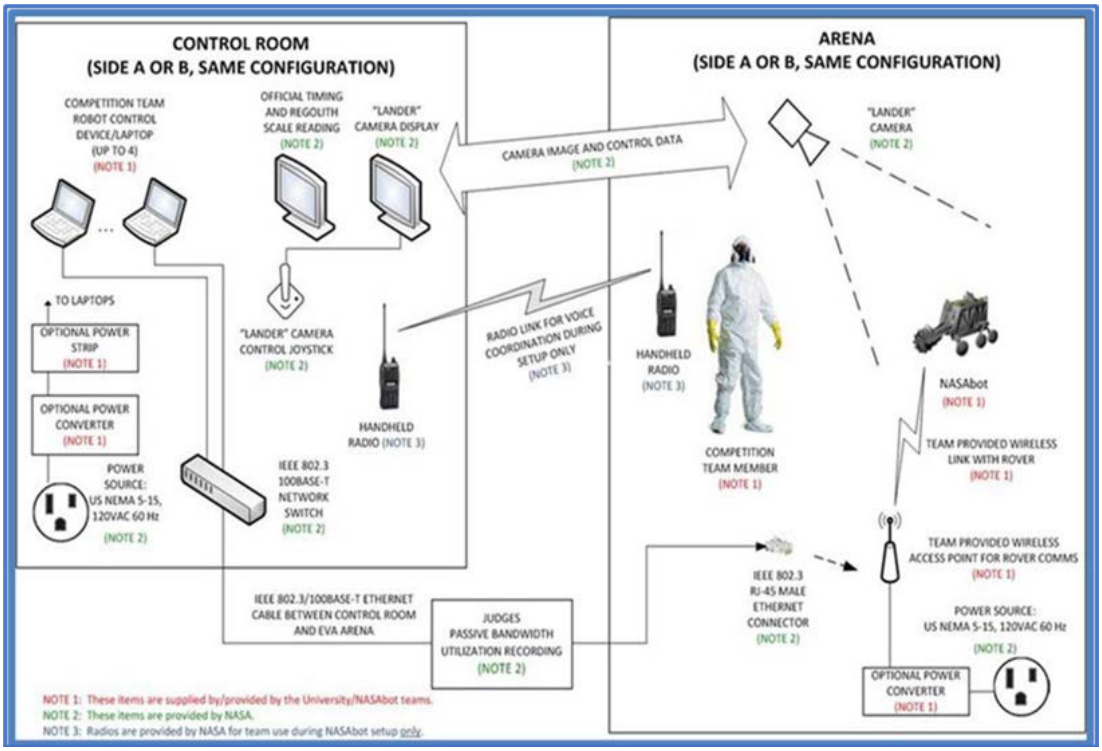


Figure 13 Comm Configuration

14.1 General Communication System Requirements

1. Teams shall use IEEE 802.11 wireless protocol standards for their wireless connection (Wireless Access Point (WAP) router and rover client).
2. Routers are required to support both the 2.4 and 5 GHz bands.
3. Routers are required to have the ability to turn off the 2.4 GHz band.
4. Each team will be assigned a Service Set Identifier (SSID) SSID upon checking in for the competition that must be used for their wireless equipment.
5. Team's SSID will be written as "Team_##."
6. Teams are required to broadcast their SSID.
7. Robots and access points are ONLY allowed to be transmitted using their designated SSID.
8. Hidden networks are not allowed.
9. Wireless encryption is required.
10. Teams cannot use channel bonding for 2.4 GHz data transmission. Meeting this rule will require a spectral mask or "maximum spectral bandwidth setting" of 20MHz for all wireless transmission equipment.

11. All team-provided wireless equipment shall operate legally within the power requirements set by the Federal Communications Commission (FCC) for unlicensed wireless equipment operating in the Industrial, Scientific, and Medical (ISM) radio frequency band. The FCC Federal Regulations are specified in the Electronic Code of Federal Regulations, Title 47, Telecommunication, Part 15, and must be followed if any commercial equipment is modified. All unmodified commercial off the shelf access point equipment and computers already meet this requirement.
12. If a team inserts any type of power amplification device into the wireless transmission system, this will likely create a violation of FCC rules and is NOT allowed in the competition.
13. The radio frequency power requirements apply to all wireless transmission devices at any ISM frequency.
14. Bluetooth transmission equipment in the 2.4 GHz range is allowed for sensors and other robot communications. Bluetooth is allowed only at power levels of Classes 2 and 3 and are limited to a maximum transmit power of 2.5 mW EIRP. Class 1 Bluetooth devices are not allowed.
15. The use of 2.4 GHz Zigbee/IEEE 802.15.4 technology is prohibited because of the possibility of interference with the competition wireless transmissions.
16. Technology that uses other ISM non-licensed radio frequencies outside of the 2.4 GHz band, such as 900 MHz or 5 GHz frequency bands, are allowed for robot or sensor systems, but these frequencies will not be monitored during the competition. Interference avoidance will be the responsibility of the team and will not be grounds for protest or rematch by any team.
17. Teams can elect to use 5 GHz Wi-Fi during competition for WAP to robot communications, however, this frequency will not be monitored for wireless interference and will not be grounds for a protest or rematch by any team.
18. Use of the NASA-provided situational awareness camera in the control room will add 500 Kbps (Kilobits Per Second) of data use for each camera. If the team elects to turn on the camera at any time during a construction run, the team will be charged for the full 500 Kbps of data use.
19. The communications link is required to have an average data utilization bandwidth of no more than 4,000 Kbps. There will not be a peak data utilization bandwidth limit.

14.2 RoboPits Communications Requirements

1. All routers in the RoboPits shall turn off 2.4 GHz by default or power down their wireless equipment.
2. The RoboPits will have full access to the 5 GHz band. This is the required Wi-Fi frequency band for use in the RoboPits if wireless connectivity is required. An alternative is to connect an ethernet cable between the devices.
3. If needed, teams in the RoboPits can test their robot using 2.4 GHz with prior authorization from the Pit Boss. These tests are to be conducted only on Channel 11 and will only be allowed for a short period of time.
4. The RoboPits communication environment will be monitored
5. If the team is found to be transmitting on Channel 1 while in the RoboPits without approval from the Pit Boss, the following will occur:

1. 1st time a warning and announcement to all teams in the RoboPits that the team was in violation.
2. 2nd time a 500-point deduction from the final score (sum of the two runs) will be assessed.
3. 3rd time will result in disqualification from the competition.

14.3 General Radio Frequency and Communications Approval (Comm Check)

1. All teams are required to successfully pass a Comm Check prior to entering the Artemis Arena.
2. Compliance with all rules specified in General Communication System Requirements Section 14.1 will be verified during the Comm Check.
3. Each team must demonstrate to the communication judges that their robot and access points are operating only on the assigned channel.
4. Each team must demonstrate wireless control of their robot.
5. Each team will have a maximum of 15 minutes at the communication judges' station.
6. Comm checks are to be performed on Channel 1.
7. Teams that are in the queue for a comm check are allowed to quickly configure to Channel 1, then turn their wireless router off until they are at the comm test station for checkout, as permitted by the Pit Boss.
8. The teams must identify and show the judges all the wireless emission equipment on the robot.
9. If the team robot is transmitting low power Bluetooth or any non-2.4 or 5 GHz frequency equipment, printed documentation from the manufacturer with part numbers of all wireless transmission equipment must be provided to the judges. This printout must be from the manufacturer's data sheet or manual, and will designate the technology, frequency, and power levels in use by this type of equipment. Teams must show that the equipment in use is not using Bluetooth Class 1 and can be verified within the documentation provided.
10. If teams are using Bluetooth on their robot, printed documentation of the hardware in use must be provided to the judges. Teams must show that the equipment in use is not using Bluetooth Class 1 and can be verified within the documentation provided.
11. The team must show that they can turn off 2.4 GHz after demonstrating control of the robot.
12. If a team cannot demonstrate the above tasks, the team will be disqualified from the competition.

14.4 Artemis Arena Communications Requirements

1. Robot operations will be performed only on Channel 1.
2. Teams must be able to configure to Channel 1 for the competition within 15 minutes of being notified to accommodate real-time scheduling changes.
3. Teams that are in the queue for a construction run must obtain Pit Boss approval to quickly configure to Channel 1, then turn their wireless off until they are notified by Arena personnel.
4. If a team is on the wrong channel during their competition runs, the team will be disqualified and required to power down.

5. Channel 1 will be monitored during competition to ensure there are no other teams transmitting on the assigned frequencies.
6. NASA-provided situational cameras are staged in the Artemis Arena. The camera joystick and camera display will be located with the team in the MCC.

14.5 Communications Artemis Arena – Mission Control Center

1. Each team will provide the wireless link (access point, bridge, or wireless device) to their robot, which means that each team will bring their own Wi-Fi equipment, router and any required power conversion devices.
2. Teams must set up their own network IP addresses to enable communication between their robot and their MCC control computers, through their own wireless link hosted in the Artemis Arena.
3. In the arena, NASA will provide an elevated network drop (male RJ-45 Ethernet plug) that extends to the Mission Control Center, where NASA will provide a network switch for the teams to plug in their laptops. The network drop in the Artemis Arena will be elevated high enough above the edge of the regolith bed wall to provide adequate radio frequency visibility of the entire arena.
4. Network connections to the robot are ONLY allowed via the WAP router and the cable going to the MCC. Any backchannel wireless connections via laptop or cellphone to the WAP router or robot during setup or competition are expressly forbidden and will result in disqualification.
5. A shelf will be set up next to the network drop at a height 0 to .5 meters above the walls of the arena and will be placed in a corner area. This shelf is where to place the Wireless Access Point (WAP) to communicate with the robot. During robot system operations during the competition there will be dust accumulation in this area.
6. NASA will provide a standard US National Electrical Manufacturers Association (NEMA) 5-15 type, 110 VAC, 60 Hz electrical jack by the network drop. This power connection will be used for the WAP router used for robot Wi-Fi 33 communications. This will be located no more than 1.5 meters from the shelf. The team must provide any conversion devices needed to interface team access points or Mission Control Center computers or devices with the provided power source.
7. Teams are strongly encouraged to develop a dust protection cover for their wireless access point (WAP) that does not interfere with the radio frequency signal performance.
8. During the setup phase, the teams will set up their access point and verify communication with their robot from the Mission Control Center.

14.6 Service Set Identifier (SSID)

A Service Set Identifier (SSID) is the unique, user-friendly name of a Wi-Fi network, used to distinguish it from others nearby. It is case-sensitive and is broadcast by routers to allow devices to identify and connect. After team selection is complete, NASA will assign team numbers.

Communications Checklist

Lunabotics Communications Inspections

Judge's Name: _____

School Name: _____

Run 1 _____ Run 2 _____ Date: _____

1. Assigned SSID - Team (School Name_##) _____
2. Team is broadcasting their SSID _____
3. Routers support both the 2.4 and 5 GHz bands _____
4. Routers can turn off the 2.4 GHz _____
5. Robots can switch between Channel 1 and Channel 11 _____
6. The team is connected to the robot by wireless remote control and can control the robot _____
7. Wireless encryption enabled _____
8. Teams cannot use channel bonding for 2.4 GHz data transmission _____

Rule Violation(s): _____

15 Mission Control Center (MCC) and Autonomous Operations Requirements

Teams will control or autonomously operate their robots remotely from the Mission Control Center (MCC) to simulate operations of a Lunar infrastructure construction mission. Mission Control Judges (MCJs) will supervise team activities in the MCC and assess their performance during each construction run. The Mission Control Director (MCD) will serve as the Lead Judge for the MCC and ensure the MCC and Autonomy rules outlined in this guidebook are interpreted and enforced consistently across all teams.

If any inconsistencies arise between section 15 Mission Control Center (MCC) and Autonomous Operations Requirements and other sections, 15 Mission Control Center (MCC) and Autonomous Operations Requirements shall prevail.



Figure 14 Mission Control Center

15.1 General Protocol

1. Each team will be allowed a maximum of 4 team members in the MCC. All members must enter the MCC together and only do so when authorized by the MCJ. Faculty and Advisors are not permitted in the MCC at any time.
2. Team members that have entered the MCC are not allowed any external communications until the completion of their run. The one exception is communication with their Artemis Arena teammates during the setup period, which is only permitted using equipment provided by Lunabotics.
3. Teams may only bring electronic devices required for robot operations into the MCC. Extraneous laptops, cell phones, smart devices, etc., are prohibited.
4. Teams are responsible for ensuring they enter the MCC with all mission-critical components and spares required that are not explicitly identified in the rules as provided by Lunabotics. Once in the MCC, team members will not be permitted to retrieve forgotten items.

5. Lunabotics provides teams with access to two situational awareness cameras in the Artemis Arena via monitors in the MCC. The MCC monitors provided for situational awareness may not be utilized by the team for any other purpose. Manipulation of NASA situational awareness cameras, if in use, is permitted during autonomy attempts.
6. Teams must resolve all questions and rule clarifications pertinent to a construction run before entering the MCC for that run. The competition schedule will not be delayed accommodating last-minute requests of this nature.
7. MCJs are observers only and are not allowed to provide “help” during robot operations. Mid-run questions, such as whether the robot is in an acceptable position or if certain points have been attained, will not be answered.
8. It is the sole responsibility of team members in the MCC to communicate effectively with the MCJ to ensure every autonomy attempt is recognized. If the MCJ is not sufficiently notified of the attempt in advance of the team initiating its execution or notified of its completion prior to the team resuming manual control, the attempt will not earn any points.
9. Telerobotic operators are only allowed to use data and video originating from the robot and the competition video monitors.
10. The Mission Control and Arena judges have the authority to terminate a setup period or construction run at any time if the team is not using them in accordance with the rules.
11. Teams are expected to conduct themselves in a professional manner as if executing a NASA operation.
12. Teams are expected to use sound engineering practices and principles to operate their robot.
13. Team members must comply with all directions from the MCJ.
14. Disputes with MCJ direction or decisions must be elevated through the MCD.

15.2 Setup Protocol

1. Teams may not connect or interact with any equipment in the MCC until the setup period begins.
2. The setup period is for placing the robot in the Artemis Arena and bringing it online for the construction run. Teams are allowed up to 10 minutes to connect their laptops and routers, establish communications with their robot, and perform any initial systems checkout required. Teams must indicate competition readiness to the MCJ as soon as their robot is ready.
3. During the setup period, the MCJ will provide the team with a handheld radio to enable communications with team members in the Artemis Arena. Teams must return this radio to the MCJ at the end of the setup period and it may not be used during the construction run.
4. Arena team members are prohibited from informing or pointing out obstacles, craters, and other arena conditions to the MCC team members.
5. Teams may use the situational awareness cameras during the setup period without penalty.

15.3 Anomaly Protocol

1. Once the competition timer has started, the robot has 5 minutes to move on the mission. If the robot has not initiated a mission cycle by the 5-minute mark, the robot is considered inoperable, and the run will end.
2. As responsible engineers, the team should notify the MCJ that they are ending the run if their robot experiences an unrecoverable issue that renders it incapable of executing key mission tasks. Such failures include:
    - Loss of Comm: The robot is functional but can no longer communicate properly with the MCC.
    - Loss of Locomotion: The robot ceases movement or experiences infrequent, non-continuous movements for a period of 5 minutes.
    - Loss of Excavation: The robot can no longer acquire regolith per its design.
    - Loss of Deposition: The robot can no longer offload regolith per its design.
    - Loss of Robot: The robot is fully unable to execute the mission. This scenario could be due to technical issues or unfavorable interactions with the competition arena.
3. In the event a robot experiences a mission-ending anomaly and the team does not voluntarily end the run within a reasonable amount of time, Mission Control and Artemis Arena judges have the authority to terminate the run. “Reasonable” is at the judge’s discretion based on the specific circumstances of the run. Teams “joyriding” or otherwise wasting competition time may be assessed penalty points.
4. It is the team’s responsibility to ensure they are executing corrective actions efficiently and communicating properly with the MCJ about long cycle steps, such as full system resets, that will make the robot appear further inoperable. Failure to do so could result in the termination of the run.

15.4 Post-Run Protocol

1. Teams must cease operations when the competition timer ends.
    - If the robot is in the middle of an autonomous activity, teams must send a command to inhibit their robot from taking any further actions.
    - Regolith offloading is permitted to be completed only if the robot was actively dumping material prior to the expiration of the competition timer.
2. Teams may not disconnect communications with their robot or begin dismantling their MCC equipment until directed to do so by the MCJ. Sustained operability is necessary in the event the robot must relocate or unload regolith prior to its removal from the Artemis Arena.
3. Teams must remain in the MCC until dismissed by the MCJ.
4. Teams are responsible for ensuring they leave the MCC with all equipment they have brought into it. Once the next team has entered the MCC, forgotten items cannot be retrieved until that team’s run is complete.

16 Autonomous Operations

During each competition run, the team will earn up to 1400 Construction points for autonomous operation. As MCJs are not intimately familiar with each robot's concept of operations (ConOps) procedures, it is the sole responsibility of the team members in the control room to coordinate with and inform the MCJ of each attempt for autonomy points to make sure their autonomous attempts are recognized and therefore scored correctly.

A tie in autonomy points relative to the Caterpillar Autonomy Award will be broken based on the total construction score (sum of both runs).

Construction points will be awarded based on the percent of the total cycle segments made in automation/autonomy. The segments of a complete cycle are excavation, travel loaded, dump, travel (return) empty – repeat. Based on a robot's design there can be many excavation cycles for one travel/dump cycle. Previous year's rules rewarded a team's ability to demonstrate automation/autonomy once in each cycle segment– excavation, travel, dump. However, this is not representative of what will be required in lunar excavation based on the latencies involved in direct remote control. This year's rules are intended to encourage teams to use automation/autonomy at a much greater percent of their overall operations during any run. This will require a high level of robustness in the automation/autonomy capability.

Teams must complete an autonomy review with autonomy judges prior to their first run. In this review they must walk through the autonomy system and how it is implemented. In this review judges will determine if the team's approach meets the defined, allowed use of arena walls. The judges will also be determining if the autonomy approach can achieve travel autonomy without the use of "bread crumbs". Teams must demonstrate their visualization of the real time obstacle detection and associated mapping of obstacles and the resulting path planning that will be required in Mission Control – see Section 15 Mission Control Center (MCC) and Autonomous Operations Requirements. If teams cannot explain their system to the satisfaction of the judges, then they will not be eligible for Autonomy points.

16.1 Excavation Automation: Maximum 200 pts

1. The Excavation Zone overlaps with the Starting Zone. Teams may excavate material anywhere in the Excavation, which includes the Starting Zone.
2. Within the Excavation Zone, the team must indicate to the MCJ that they are going hands-free for the excavation attempt.
3. The robot must execute machine control commands during the excavation task.
4. The robot must demonstrate the ability to excavate and collect regolith for transport. A discernable amount of regolith must be collected for transport as determined by the MCJ. The MCJ may engage the arena judges for confirmation if camera angle or performance does not allow confirmation in Mission Control. Collection can be in an internal hopper, a bucket, a blade, a bucket drum, etc. – based on the robot design. Hands-free operation must begin before the robot engages the regolith to begin the excavation process.
5. Excavation mechanisms must be completely removed from contact with the regolith before returning to remote control operation.
    a. The team may remain in hands free operation and go directly into travel loaded automation.
6. Once excavation is complete the team must indicate they are going to remote control before taking control of their robot.

7. This level of automation will require teams to master the lower-level machine control of their robot platform associated with excavation. It is noted that past teams have proven this capability to be helpful in achieving better excavation results, as the coordination of human commands for the robot for excavation can be difficult to master.

16.2 Dump Automation: Maximum 100 pts

1. Teams are allowed to traverse the Obstacle Zone via remote control.
2. Prior to crossing into the Construction zone, they must indicate to the MCJ that they are going hands-free for the dump attempt.
3. The robot must execute machine control commands during the dump task.
4. The robot must place the regolith at the berm construction location. A discernable amount of regolith must be placed at the berm location as determined by the MCJ. The MCJ may engage the arena judges for confirmation if camera angle or performance does not allow confirmation in Mission Control.
5. Once dumping is complete the team must indicate they are going to remote control before taking control of their robot.
    a. The team may remain in hands free operation and go directly into travel empty automation.
6. This level of automation will require teams to master the lower-level machine control of their robot platform associated with dumping. In addition, teams will need to master localization in the construction zone as well as path planning to align and place regolith at the designated berm construction location.

16.3 Travel Automation: Maximum 500 pts

1. The competition judges will attempt to construct the obstacle field in such a way as to require obstacle detection, mapping, and navigation planning to determine a “slalom” route to reach the construction zone. The teams shall not architect a “Point and traverse” approach for this automation step.
2. Walls may not be used for the purpose of mapping autonomous navigation and collision avoidance as there are no walls on off-world locations. Touching or having a switch sensor spring wire that may brush on a wall or any other surface as a collision avoidance sensor is not allowed Touch sensors are also prohibited. The real intent of this Autonomy Rule is to ensure that a priori knowledge (e.g., dimensions and location relative to the operational area of the arena) of the walls and/or the column are not being used for localization. This means that the walls and/or column, when detected, can be used as features for localization without a prior knowledge. In addition, a simple tracking offset cannot be used. For example, following the wall 0.5m away to traverse from one end of the arena to the other would not be allowed.
3. The point penalty associated with “bread crumbs” or defining waypoints via remote control has been removed for 2027. Teams attempting travel automation must have a visualization of the robot’s real time obstacle detection, associated mapping of obstacles, and resulting path planning for MCJ verification that “bread crumbs” are not being used. No real time interaction to control or influence the robot can be utilized in this system.
4. Travel Loaded

   a. The teams must indicate to the MCJ that they are going into hands-free mode while still in the excavation zone. The robot must remain in hands-free mode while crossing the obstacle field and crossing into the construction zone. This level of automation will require the team to master the following:

   b. Localization across the entire competition arena

   c. Object detection and location relative to the robot

   d. Navigational planning based on location and obstacles/traversable areas

5. Travel Empty

   a. The teams must indicate to the MCJ that they are going into hands-free mode while still in the construction zone. The robot must remain in hands-free mode while crossing the obstacle field and crossing into the excavation zone. This level of automation will require the team to master the following:

   b. Localization across the entire competition arena

   c. Object detection and location relative to the robot

   d. Navigational planning based on location and obstacles/traversable areas

16.4 100% of Automation of Cycles 1000 pts

1. It is possible a team may complete all of their excavation, travel, and dump cycles in automation, but still have some remote control “stick time” during their run. If that combination occurs the team will achieve 1000 points for Autonomy.

16.5 Full Autonomy 1400 pts

1. Teams may begin in remote control and move the robot within the starting zone only to localize.
2. Teams must begin with hands-free control while still in the starting zone and remain in hands free mode for the entirety of the rest of the construction run.
3. At least two cycles of excavating, crossing the obstacle field, crossing into the construction zone, dumping the regolith for the berm, crossing the obstacle field and returning to the excavation zone must be completed during the construction run.
4. Berm construction (volume), as determined by the volumetric scan, must be achieved for this level of autonomy.
5. This level requires mastery of all aspects of autonomy associated with this competition and demonstrates a level of robustness to complete at least two full cycles. System robustness is essential for terrestrial and extra-terrestrial construction.

16.6 Autonomous Scoring

1. % Excavation Automation: Points for excavation will be based on how many total excavation cycles the robot executes during the run, and how many of those cycles were successful in automation per the above rules.
2. % Dump Automation: Points for excavation will be based on how many total excavation cycles the robot executes during the run, and how many of those cycles were successful in automation per the above rules.
3. % Travel Automation: Points for travel will be based on how many total travel cycles (loaded + empty) the robot executes during the run, and how many of those cycles were successful in automation per the above rules.

16.6.1 Berm Volume Multiplier

The purpose of this factor is to encourage teams to still build reasonable berms (the task) and to discourage a team from running only one cycle for each segment – achieving the 1000 autonomy points, while building a very small berm. To implement this a berm volume is applied. This threshold is 25 liters of berm volume or 25,000 cm^3. Above the 25,000 cm^3 level the berm multiplier is a factor of 1.0. Below the 25,000 cm^3 level the berm multiplier is calculated as berm volume cm^3 / 25,000 cm^3. This also means that if no berm is built (0 $cm^{3)}$ then no autonomy points will be awarded, even if the robot demonstrated autonomy capability.

16.6.2 Scoring Formula

IF Full Autonomy, THEN Autonomy Score = 1400 * Berm Multiplier

ELSE

IF (%Excavation Automation + %Dump Automation + % Travel Automation) = 300%, THEN Autonomy Score = 1000 * Berm Multiplier

ELSE

Autonomy Score = (%Excavation Automation * 200 + %Dump Automation * 100 + %Travel Automation * 500)*Berm Multiplier

16.6.3 Example of Scoring

Example 1: The following table assumes that for a given run the team executed 8 excavation cycles, 15 travel cycles (8 loaded + 7 empty) cycles, and 8 dump cycles. The team was able to execute automation for each of excavation, travel, and dump one time during the attempt. The team achieved 50,000 cm^3 of berm volume.

Table X1: Example 1 cycle of automation for each cycle segment.

Cycle Segment	Total Cycles	Cycles in Autonomy	% in Autonomy	Max Autonomy Points	Autonomy Points
Excavation	8	1	12.5	200	25
Travel	15	1	6.7	500	34
Dump	8	1	12.5	100	13
Total					72

Berm Multiplier = If berm volume => 25,000 cm^3, then multiplier = 1.0, else multiplier = berm volume cm^3/ 25,000 cm^3.

Final autonomy score = 72 * 1.0 berm multiplier = 72

Example 2: The following table assumes that for a given run the team executed 2 excavation cycles, 3 travel cycles (2 loaded + 1 empty) cycles, and 3 dump cycles. The team was able to execute automation for each of excavation, travel and dump all cycles during the run. However, if the team used remote control at some point during their run, the run does not qualify as fully autonomous. This means they cannot earn the 1400 points awarded for full autonomy, and their score will instead be calculated using the formulas provided in this section.

The team achieved 12,500 cm^3 of berm volume.

Table X: Example All cycles in automation for each cycle segment.

Cycle Segment	Total Cycles	Cycles in Autonomy	% in Autonomy	Max Autonomy Points	Autonomy Points
Excavation	2	2	100	200	
Travel	3	3	100	500	
Dump	2	2	100	100	
Total					1000

Berm Multiplier = If berm volume => 25,000 cm^3, then multiplier = 1.0, else multiplier = berm volume cm^3/ 25,000 cm^3. 12,500 cm^3/ 25,000 cm^3= 0.5 berm multiplier

Final autonomy score = 1000 * 0.5 berm multiplier = 500

17 Scoring

The mission of this robot system is to construct a berm that protects astronauts and habitat modules from plume ejecta from vehicles lifting off and landing on the opposite side of the constructed berm. For this competition, plume ejecta is presumed to move perpendicular to the surface and parallel to the short axis of the berm area. Both mass (or volume) of the berm and the projected area of the berm normal to the plume ejecta provide this protection.

The berm construction final score will be cumulative, not the highest of the two runs, and not an average of the two runs.

17.1 Construction Points

The robot must build a measurable cm^3 of berm to achieve Construction Points. If the berm volume is 0 cm^3 then the Construction Points will be zero.

1. **Pass All Inspections (Comm and Safety) -** each team is required to perform a safety inspection and communications check prior to the first construction run. This should be performed as early as possible after check-in to ensure compliance with all rules and communication functionality. Neither is optional, if one cannot pass, the robot will not be permitted in the Arena. A safety inspection will also be conducted prior to each construction run. There are no points associated with the inspections, they must pass to proceed.

2. **Construction Berm Productivity – Normalized for Robot Mass (BP Mass)** A volumetric scan of the berm target area will be performed before and after the run. Only the berm volume of portions of the constructed berm within the target area for berm placement will be counted. Reference section 11.1 Arena Composition. There is no restriction on the shape, height, or number of berms constructed. The team will earn construction points for each cubic cm of berm constructed above grade per minute per each kg of the robot's mass **(**cm^3 **berm / min / kg robot mass).**

3. **Construction Berm Productivity – Normalized for Energy (BP Energy)** The team will earn construction points for each cubic cm of berm constructed above grade per each watt hour of energy consumed as displayed on the COTS electronic data logger and verified by an arena judge **(**cm^3**berm /min /watt hour)**. **Penalty** - A 30 point penalty will be accessed if the robot energy data logger is not wired in accordance with the Guidebook.

4. **Bandwidth Use** - During each competition run, the team will be scored on bandwidth usage as follows: (# arena cameras used x 0.5) + Average Bandwidth (Mbps) = Total Bandwidth (Mbps). If Total Bandwidth (Mbps) > 4 = 0, else = 120 – (Mbps x 30).
    i. Example: (1) arena camera used, 0.5 Mbps average bandwidth. (1x0.5) + 0.5 = 1.0 Total Mbps
    ii. 120 - (1.0 x 30) = 90 points

5. **Dust Tolerant Design** - During each competition run a team can earn up to 30 points for dust tolerant design features on the construction robot. If the construction robot has exposed mechanisms where dust could accumulate during a lunar mission and degrade the performance or lifetime of the mechanisms, then fewer Construction points will be earned in this category. Teams are encouraged to point out dust tolerant features to the judges during inspection. The judges will allocate these points based on an inspection for the construction run. All decisions by the judges regarding dust tolerance are final. The points for dust-tolerant design are as follows:

    a. Drive train and components enclosed/protected: 15 points

    b. Active dust control (brushing, electrostatics, etc.): 5 points

    c. Custom dust sealing features (bellows, seals, etc.): 10 points

6. **Dust Free Operation** - During each competition run, a team can earn up to 30 points for dust free operation. The judges will allocate these points based on actual performance during the competition runs. If the construction robot projects or raises a substantial amount of airborne dust due to its operations, fewer construction points will be earned in this category. Ideally, the construction robot will operate in a clean manner without dust projection, and all mechanisms and moving parts will be protected from dust intrusion. All decisions by the judges regarding dust projection are final. The points for dust-free operation are as follows:

    a. Driving without dusting up regolith: 5 points

    b. Digging without dusting up regolith: 20 points

    c. Transferring regolith without dumping the regolith on the robot: 5 points

7. **Autonomy** – Reference section 16 Autonomous Operations

8. **Obstacle Penalty -** If the robot comes in contact with a rock or drives across a crater, as determined by the MCJ or Arena judges, a 50-point reduction for the first occurrence will be applied, with 25 points applied for additional occurrences up to a maximum of 100 points. This is only true in

the obstacle zone. The robot is allowed to move rocks and fill in craters in the excavation and construction zones.

9. Total Points

Numbers from score sheet Example:

Robot Mass: 66kg

Berm Volume: 77551 cm^3

Energy used: 36Whr

cameras used: 1

Average Bandwidth used: 0.7 Mbps

Dust Tolerant Score: 30

Dust Free score: 25

Autonomy: 75

Penalty: -50

Construction Category Elements	Scoring Formula	Raw Value	Example	Example
1. ***Pass All Inspections (Comm and Safety)***.	NA	NA	NA	**Go Run**
2. ***Berm Construction Productivity Normalized for Robot Mass (BCP Mass)***	4.4 x cm^3 berm / min / kg robot mass	77551	4.4 x 77551 cm^3/ 15 min run/ 66 kg	**344.6**
3. ***Berm Construction Productivity Normalized for Energy (BCP Energy)***	1.5 x cm^3 berm / min /watthour	77551	1.5 x 77551 cm^3/ 15 min run/ 36 Whr	**215.4**
4. ***Bandwidth Use***	# Cameras Avg bandwidth Mbps	1 camera 0.7 avg bandwidth Mbps	TAB = (1x0.5)+0.7 = 1.2 120 - (1.2 x 30) = 84	**84**
5. ***Dust Tolerant Design***		30	30	**30**
6. ***Dust Free Operation***		25	25	**25**
7. ***Autonomy***	Task	See section 16 Autonomous Operations	75	**75**

8. ***Obstacle Penalty***		-(0 to 100)	-50	**-50**
9. ***Total Points***				**724**
10. ***Total Construction Points***			If berm volume > 0 cm^3 = Total Points, ELSE = 0 points	**724**

17.2 Artemis Grand Prize Score

1. Project Management Plan (PMP) will contribute up to 10 points towards the grand prize score.
2. Systems Engineering Paper (SEP) score will contribute up to 25 points for the grand prize score.
3. Berm Construction (BC) score will be normalized to the maximum Berm Construction points achieved by all teams to a scale up to 40 points for the grand prize

18 Rubrics

18.1 General Submission Requirements

Deliverables must be submitted in the specified file format, with the specified file naming convention, by **noon Eastern Time** on the due date. All deliverables will be submitted at the AMF Lunabotics Challenge website. Late deliverables are not accepted unless the official guidebook or official FAQ states otherwise.

The cover or title page of each rubric shall include the signature of the faculty advisor and a statement that they have read and reviewed the paper prior to submission to NASA. The cover page shall include the title of paper, full names of all team members, and university name.

18.2 Rubric - Project Management Plan (PMP)

18.2.1 PMP Resources Available

Project Management and Systems Engineering educational videos can be viewed at **NASA Lunabotics Videos**.

These videos include a series titled ***Systems Engineering for University Level Engineering Competitions Videos*** that introduces key project management and systems engineering concepts and methods useful for the Lunabotics Challenge.

Also included are **seven** different project management and systems engineering **seminars** that were recorded live at Lunabotics Challenges at Kennedy Space Center. The most immediately relevant of these for the Project Management Plan is the seminar titled ***How to Develop the Perfect Lunabotics Systems Engineering Paper: And Win the Whole Competition***. We recommend viewing this video at the very start of the project.

18.2.2 Directions

Each team shall submit a complete Project Management Plan (PMP) electronically in one PDF file. include the signature

This is an initial plan. As the project is executed, things will change and the project will evolve, which is normal and expected. In the Systems Engineering Paper discuss the changes to the plan and how the project adapted.

The cover page shall include the signature of the faculty advisor and a statement that they have read and reviewed the paper prior to submission to NASA. The cover page shall include the title of paper, full names of all team members, and university name.

The maximum length of the plan is 5 pages and the cover page will not count towards the page limit.

Any content over 5 pages will not be judged and will not be scored.

Format: The Project Management Plan shall be formatted professionally, organized clearly so that each required rubric element is easy to find, with correct spelling and grammar, with text no smaller than size 12 point font in the main body, with text no smaller than size 9 point font in graphics and tables, and using professional margins.

18.2.3 Scoring Rubric - Project Management Plan (PMP) – Worth up to 10 Points

18.2.3.1 Element 1 - Initial Project Schedule – Worth 3 Points

1. Provide a Gantt Chart or equivalent that shows the project's major due dates and events to include at least the items listed below. Discuss these only as needed.
    a. Start Date
    b. Completion Date (after project decommissioning; this is the date when the robot system will be disposed after the challenge; for example, the system is handed over to next year's team, disposed of, or other)
    c. Dates for the Major Review milestones: as a minimum, these shall include:
        i. Systems Requirements Review
        ii. Preliminary Design Review
        iii. Critical Design Review
        iv. Others may be identified as found appropriate
    d. Product delivery dates to the Lunabotics Engineering Challenge, including as a minimum:
        i. Delivery of Systems Engineering Paper
        ii. The planned date to submit Proof of Life Video
    f. Optionally, it is acceptable to also identify any major Systems Engineering activities in the Initial Project Schedule.

2. If any critical milestones are identified in the discussion of the *Initial Project Cost Budget* (Element 2 – Initial Project Cost Budget), show the dates for these milestones related to the Project Cost Budget.

3. If any critical milestones are identified in the discussion of *Initial Technical Performance Measures* (Element 4 - Initial Technical Performance Measures), show the dates for these milestones related to the Technical Performance Measurement budget.

4. Discuss how the evolution of the schedule during the life of the project will be managed. How often and when plan to review the project schedule, and the plan for how to adapt to schedule slips or schedule advance opportunities.

18.2.3.2 Element 2 - Initial Project Cost Budget – Worth 3 Points

1. Provide an estimate of the Total Project Cost, inclusive of all possible costs throughout the project life cycle, independent of any funding sources.
2. Provide a Table of Major Budget Categories and Items including the following list items as a minimum. Discuss only as needed.
    a. Breakdown of total project cost estimate for at least the following major items. Total should add up to the estimate of the Total Project Cost.
    b. Cost estimates for elements in the earliest level System Hierarchy (provide this earliest level System Hierarchy, which may only be two levels at this early point in the project). Break these estimates down as follows.
        i. Labor costs, if any
        ii. Material costs for production and completion of Lunabotics deliverables
    c. Travel costs including costs to travel to KSC (and elsewhere) if intended to accept a proffered invitation to participate in NASA's Lunabotics On-Site Challenge
3. Identify any Critical scheduling milestones and dates for budget or cost related items. These critical cost budget milestones must be included in item 5 for the *Initial Project Schedule* (Element 1 - Initial Project Schedule). For example, dates funds will be needed, dates of planned activities to raise funds, or others.
4. Discuss how the evolution of the budget during the life of the project will be managed (how often plan to review the project budget and when, and the plan for how to adapt to budget shortfalls or possible cost savings should they occur).

18.2.3.3 Element 3 - Project Technical Objectives – Worth 1 Point

List and discuss the specific technical criteria or characteristics that the team intends to achieve in the system design and operations to win the competition.

Typically, these are some measurable technical parameters to minimize or maximize, increase to some limit, or decrease to some limit. These provide the technical direction and focus for the entire team in

design and operations. For example: the team might choose to minimize mass, or maximize automation, or minimize bandwidth, or some combination of minima or maxima of these or any other criteria, to produce a system that will win the competition.

18.2.3.4 Element 4 - Initial Technical Performance Measures (Technical Budgets) – Worth 3 Points

A Technical Performance Measure is any quantifiable technical characteristic or parameter that the team may consider important, difficult to achieve, or particularly risky to project success. These may derive from the Project Technical Objectives. For example, it may be that total mass (high or low) is important, or that a low bandwidth is difficult to achieve, or that high speed is risky for the project, or any number of other quantifiable and measurable technical quantities.

1. Provide a Table of Technical Performance Measures that are deemed important for the design to accomplish the mission.
2. Include the following in the table as a minimum. Discuss only as needed.
    a. Identification of the Technical Performance Measures chosen to monitor and manage
    b. Initial Target Value for each Technical Performance Measure to be achieved in the system by the challenge

1. Provide the allocation of each Initial Target Technical Performance Measure value across the elements of the earliest System Hierarchy at each level (may only be two levels at this early point in the project; should combine at lower levels to the total Target at the highest level in the hierarchy).
4. Discuss any critical schedule milestones and dates for achieving critical technical performance levels. For example, decision points in the design process where if unable to achieve the desired level would change the design. These must be included in item 6 for the Initial Project Schedule.)
5. Discuss how the evolution of the Technical Performance Measurement budgets during the life of the project will be managed. How often the team will plan to review current Technical Performance Measure values, and the plan for how to adapt to performance shortfalls or exceedances should they occur.

18.2.4 Tips on the PMP

1. Make sure the Gantt Chart (if using) and tables are readable if provided to satisfy a rubric element. If the judges can't read it, the judges can't award points. If a graphics is provided with unreadable (less than 9-point text as viewed in the pdf) text, make sure the information to satisfy the rubric elements are discussed in the main body (at least 12-point font).
2. Initial Project Schedule: Major Reviews. Make sure at least 3 show up on the schedule: SRR, PDR, and CDR. If using an alternate review instead of SRR, PDR, and CDR, then identify in the discussion the name of the review that replaces each. Also provide the entry and exit criteria of the alternate review.
3. Developing any management plan boils down to two things: making decisions and writing them down. It's important to document those decisions at the beginning of the project and share with the

team. Update the plan as more is learned and the project progresses. The Project Management Plan rubric identifies an important subset of the decisions to be made in the Lunabotics project planning process, and simply requests the team to provide those specific decisions.

4. Initial Project Budget: Budget evolution. As the design matures, the team will learn more about how much the project is to actually going to end up costing. To be sure the team doesn't run out of money before the end of the project, set up periodic reviews of the cost budget. In case the costs are much greater (or much less) than expected, decide at the beginning of the project how to address budget shortfalls, and what the team will do with budget excesses. Remember to make these decisions and discuss them in the PMP.

5. Initial Technical Performance Measures (TPM): Allocation to System Hierarchy. The system hierarchy is the backbone of the project. At the very start of a project, it may only consist of two levels. That early decomposition allows the team to allocate how much time to plan to spend on each sub-element in the hierarchy, how much money the team plans to spend on each sub-element, and how much of the technical performance measure's values each sub-element will have to provide so the system can accomplish the mission. As the design matures, these early allocations of TPM values across the system hierarchy levels will guide the team through the design process to deeper levels in the system hierarchy, and enable sub-allocations of TPM values across these new levels. And of course, the team may learn things in the design process that might change even those earliest allocations of TPM values. There is an example of allocations down through the system hierarchy in ***SE Video 2: The Central Elements of Project Management***. That video describes budget allocation across the system hierarchy; the approach is similar for mass and other TPM allocations. Don't forget to include the TPM allocations in the PMP.

18.3 Rubric - Preliminary Design Review Report- Worth up to 100 Points

This report provides products for a successful PDR that produces the Allocated Baseline.

The Allocated Baseline is produced by an iterated process of Functional Analysis and Decomposition of the functions and performance required to accomplish the Lunabotics mission. The Allocated baseline consists of descriptions (as sets of requirements only) of subsystems, assemblies, and possibly subassemblies for which the final detailed design (as design drawings, circuit schematics, material specifications, pseudocode, etc.) can be completed within single traditional engineering disciplines (including software engineering) for each.

The Preliminary Design Review Report shall be provided in single pdf format file. The cover page shall include the signature of the faculty advisor and a statement that they have read and reviewed the report prior to submission to NASA. The cover page shall include the title of report, full names of all team members, and university name.

18.3.1 Allocated Baseline Elements - Worth up to 60 Points

The first four required report elements provide an overview of and insight into the Allocated Baseline. These four elements shall be judged as a whole, based on five characteristics:

1. The assessment of the overall system's ability to enable the accomplishment of the Lunabotics mission
2. Completeness of the design at this level
3. Consistency of the design elements (all at the appropriate level of detail)
4. Validity of the design at this level (addresses the problem the Lunabotics Mission is to solve)

5. Coherency of the design at this level (lack of any internal conflicts)

18.3.1.1 Scoring Element 1: Allocated Baseline System Hierarchy

The paper size for this graphic may be any American paper size up to ANSI D (34” x 22”), and the type size shall be no smaller than 12 points at full scale. This element shall be provided on one page only.

There shall be no discussion provided for this report element.

1. Provide the complete system hierarchy baselined at PDR. The elements on the bottom row shall be all those elements baselined at PDR.
2. Provide on this graphic the TPM allocations for every hierarchy element at all levels.

18.3.1.2 Scoring Element 2: Function and Discipline Allocation Matrix

This element shall be on ANSI A paper (8.5” x 11”) and all text for it shall have a type size of 12 points. The matrix may use as many pages as necessary to address every Allocated Baseline System Hierarchy element.

There shall be no discussion provided for this report element.

1. Column 1 shall list the hierarchy elements
2. Column 2 shall list the independent functions the hierarchy element must provide to accomplish the Lunabotics mission (there may be multiple functions that one element must provide, but one function cannot be provided by multiple system hierarchy elements)
3. Column 3 shall provide the one traditional engineering discipline to which the element was allocated (one of: Mechanical Engineering, Electrical (wiring, motors, actuators, etc., i.e., not employing solid-state electronics), Electronic (where solid-state electronics are used, e.g., processors, sensors, communications, etc.), or Software Engineering).

18.3.1.3 Scoring Element 3: Hardware Conceptual View

The paper size for this element may be any American paper size up to ANSI D (34” x 22”), and the type size shall be no smaller than 12 points at full scale. This element shall be provided on one page only.

There shall be no discussion provided for this report element.

Provide a notional breakout view of the system showing all of the Allocated Baseline hierarchy elements with notations identifying each system hierarchy element. This may require multiple views (i.e., from different angles, or internal and external views) to show all of them. Each Allocated Baseline system hierarchy element shall appear in at least one of the views.

18.3.1.4 Scoring Element 4: Concept of Operations

The paper size for this element may be any American paper size up to ANSI D (34” x 22”), and the type size shall be no smaller than 12 points at full scale. This element shall be provided on one page only.

There shall be no discussion provided for this report element.

1. Provide a graphic showing the Concept of Operations baselined at PDR. This Concept of Operations consists of all activities where the team members use the system hierarchy elements to accomplish the Lunabotics mission. It shall begin whenever the final testing is complete and successful at the site, and show all operations starting with packing up and delivering the system to the competition, throughout the entire competition including in the arena, and completing with final disposal of the system (trashing it, handing it off to next year's team, etc.).
2. Provide a timeline graphic for all team operations in the Concept of Operations.

18.3.2 Scoring Element 5: System Hierarchy Specifications- Worth up to 40 Points

This element shall be on ANSI A paper (8.5" x 11") and all text for it shall have a type size of 12 points. This report element may use as many pages as necessary.

There shall be no discussion provided for this report element.

Each specification shall be scored based on the judge's perceived ability as a traditional discipline engineer to use the specification to complete the final design (down to subassemblies, components, parts, as detailed design drawings, schematics, material specifications, pseudocode, etc.).

Scoring of each of the four specifications shall be on a scale of 0-10 points for a maximum of 40 points for this report element.

This element is a set of four specifications for Allocated Baseline system hierarchy elements (subsystems, assemblies, subassemblies that appear on the bottom level of the System Hierarchy). These specifications (sets of requirements) should be sufficiently complete such that a single traditional engineering discipline can produce the final detailed design (engineering drawings, material specifications, electronic schematics, pseudocode, etc.).

1. Select an Allocated Baseline hierarchy element assigned to the Mechanical Engineering discipline and provide its complete, consistent, valid, and coherent specification (set of requirements) as baselined at PDR for critical detailed final design.
2. Select an Allocated Baseline hierarchy element assigned to the Electrical Engineering discipline (power, wiring, harnesses, batteries, motors, actuators, and other non–solid-state functions) and provide its complete, consistent, valid, and coherent specification (set of requirements) as baselined at PDR for critical detailed final design.
3. Select an Allocated Baseline hierarchy element assigned to the Electronic Engineering discipline (solid-state electronics such as processors, sensors, and communications) and provide its complete, consistent, valid, and coherent specification (set of requirements) as baselined at PDR for critical detailed final design.
4. Select an Allocated Baseline hierarchy element assigned to the Software Engineering discipline and provide its complete, consistent, valid, and coherent specification (set of requirements) as baselined at PDR for critical detailed final design.

18.4 Rubric - Systems Engineering Paper (SEP)

18.4.1 SEP Resources Available

Project Management and Systems Engineering educational videos can be viewed at **NASA Lunabotics Videos**.

These videos include a series titled ***Systems Engineering for University Level Engineering Competitions Videos*** that introduces key project management and systems engineering concepts and methods useful for the Lunabotics Challenge.

Also included are **seven** different project management and systems engineering **seminars** that were recorded live at Lunabotics Challenges at Kennedy Space Center. The most immediately relevant of these for the Project Management Plan is the seminar titled ***How to Develop the Perfect Lunabotics Systems Engineering Paper: And Win the Whole Competition***. We recommend viewing this video at the very start of the project.

Undergraduate course materials in systems engineering (one example) are available Here.

18.4.2 Directions

1. Each team shall submit a complete Systems Engineering Paper electronically in one PDF file and include the Faculty signature, and School Name.
2. The purpose of the Systems Engineering Paper is for the team to demonstrate how they used the systems engineering process in designing, building, and testing their system. **At least one explicit, clear, and preferably numerical example should be provided for every assertion** related to elements of the Systems Engineering Paper Rubric.
3. A minimum score of 20 out of 25 possible points must be achieved to qualify to win in this category. In the case of a tie, the judges will choose the winning Systems Engineering Paper.
4. Either one-column or two-column formatting is acceptable. The “professional journal margins” requirement reminds authors to provide appropriate white space at the margins and between sections.
5. Deliverables such as the Systems Engineering Paper (SEP) must comply with expectations of academic honesty at the host university. Plagiarism concerns are taken seriously and are grounds for disqualification.
6. Teams are reminded that plagiarism of a team’s previous Systems Engineering Paper is still plagiarism. The content for this SEP may be inspired by SEPs from previous years’ competitions, but the exact content must be developed specifically for this delivery.

Note to Student Teams:

- The phrase “significant requirement” is used in this SE Paper Rubric, replacing “key driving” requirements used previously. The intent is the same. Unlike formal project deliverables, a page limit is imposed for this deliverable. The 25-page limit on the SE Paper may be too short for the team to present all of the requirements. The teams are requested to demonstrate the applied systems engineering to the project through a sampling of the requirements.
- At some NASA centers, the term “Key Driving Requirements” is jargon and has a special and formal meaning. The intent in past competitions was for “key driving” requirements to not be the jargon term as used at some NASA centers, but to mean the requirements deemed most important for the project.

18.4.3 Scoring Rubric Systems Engineering Paper- Worth up to 25 Points

18.4.3.1 Scoring Element 1: Content – Worth 3 Points

1. Format (1 point)
    a. Provide a cover page.
    b. The cover page shall include the signature of the faculty advisor and a statement that they have read and reviewed the paper prior to submission to NASA. The cover page shall include the title of paper, full names of all team members, and university name.
    c. The Systems Engineering Paper shall consist of a maximum of 25 pages (letter size) not including the cover page, title page, table of contents, and a list of references page. If a team chooses to use appendices, they shall be included in the 25-page count. Content in the appendices shall be referenced and discussed in the main body of the paper.
    d. Only the first 25 pages of the paper will be judged.
    e. The Systems Engineering Paper shall be formatted professionally as if for submission to a professional journal:
        i. Organized clearly so that each required rubric element is easy to find
        ii. Correct grammar and spelling
        iii. Text no smaller than size 12-point font in the main body and appendices
        iv. Text no smaller than size 9-point font in graphics and tables
        v. Using professional journal margins and single spaced
2. Faculty Signature (1 point)

The cover or title page shall include the signature of the sponsoring faculty advisor and a statement that they have read and reviewed the paper prior to submission to NASA.

3. Reason for using Systems Engineering (1 point)

A statement shall be included early in the main body explaining the reason the team used systems engineering in this engineering competition (beyond that it is required). (For example: What benefit did it provide? How was systems engineering valuable to the project? There may be other reasons.)

18.4.3.2 Scoring Element 2: Project Management Merit – Worth 9 Points (1 bonus point may be awarded for exceptional work on Project Management Merit elements)

1. Is the system a New Design or Design Update? (3 points)

a. Clearly state that the system is an entirely new *clean-sheet-of-paper* system design, or that it is a substantial update to a design from a previous year or competition. Minor tweaks to a previous design, i.e., simple part replacement at the CDR level of the System Hierarchy, will not allow sufficient exercise of Systems Engineering practices to allow SE Paper scoring.

The Lunabotics System Engineering Paper should only describe the systems engineering activities performed during this competition year.

If the system design is a new clean-sheet-of-paper design, then by default the paper would address the systems engineering performed to develop all of this new design, implement it, and field it for the competition.

If the system design this year is a substantial modification of some previous system, the only systems engineering the team should describe in the SE Paper this year is the development of the modified or added system hierarchy elements (subsystems/assemblies/components), how the team interfaced the modified or added hierarchy elements with the previous system's hierarchy elements, and how the team verified that the significant requirements listed are satisfied in the modified system.

b. If the team has a substantially updated design, specifically address the following:

i. Provide the system hierarchy for the previous system and ***clearly identify*** on it which hierarchy elements were changed, added or deleted for this year's competition.

ii. Explain how the team arrived at the decisions why the team decided to change, add or delete these hierarchy elements and how they should affect the mission.

2. Major Reviews (4 points)

a. At a minimum, describe how the team conducted the **System Requirements Review** (SRR) (the control gate at which the problem to be solved stated in requirements was baselined), **Preliminary Design Review** (PDR) (the control gate at which the team baselined the requirements to be handed off to the traditional engineering disciplines for final design), and **Critical Design Review** (CDR) (the control gate at which the team baselined the final design before ordering parts or any manufacturing commences).

b. Clearly demonstrate that these reviews served as control gates, including how the team resolved the responses to comments received with the reviewers providing them.

c. Identify the external reviewers at each control gate.

d. Provide examples of comments at each Major Review received from external reviewers, and identify the reviewer who made the comment that led to changes to:

i. The system design (provide explicit, and preferably numerical, examples of requirements before the comments and the changed requirements as a result of the comments).

ii. The schedule (provide explicit, and preferably numerical, examples of schedule changes as a result of the comments).

iii. The cost budget (provide explicit, and preferably numerical, examples of cost budget changes as a result of the comments).

iv. Technical Performance Measurements (TPM) values or allocation of values across a system hierarchy level (provide explicit, and preferably numerical, examples of TPM changes as a result of the comments).

3. Schedule of work (1 point)

a. Discuss the project schedule and how it evolved from inception to disposal of robot system.

b. Provide the original planned schedule before project start (the one the team submitted as part of the PMP).

c. Provide the actual final schedule or schedule at the time the paper is delivered. The team may provide other interim schedules at relevant milestones as well.

d. Provide examples of:

i. What changed between the original and subsequent schedules. Include explicit, and preferably numerical, examples.

ii. Why it changed.

iii. When it changed.

iv. How these schedule changes produced any changes to the cost budget and any changes to technical requirements, i.e., requirements before the schedule change and the changed requirements as a result of the schedule change.

Note: When the team goes through a major review, it's not unusual for the schedule to change.

e. Demonstrate in the discussion that the schedule was used to manage the project (and not just an after the fact result of the project).

4. Cost budget (1 point)

a. Discuss the cost budget for total project costs (including travel to KSC) and how it evolved from inception to disposal of robot system.

b. Provide the original Total Estimated Project Cost and budget before project start (from the Project Management Plan).

c. Provide the actual final project cost and budget (or at the time the paper is delivered). The team may provide other interim cost budgets at relevant milestones as well.

d. Provide examples of:

i. What changed between the provided cost budgets in explicit, preferably numerical examples

ii. Why it changed

iii. When it changed

iv. How these cost budget changes produced any changes to the schedule and any changes to technical requirements, i.e., requirements before the budget change and the changed requirements as a result of the budget change.

Note: When the team goes through a major review, it's not unusual for the cost budget to change.

e. Demonstrate in the discussion that the cost budget was used to manage the project.

18.4.3.3 Scoring Element 3: Systems Engineering Merit – Worth 8 Points (up to 4 bonus points may be awarded for exceptional work and additional systems engineering methods used)

If the system design is a new clean-sheet-of-paper design, then by default the System Engineering Merit Elements below should address the systems engineering the team performed to develop all of this new design, implement it, and field it for the competition.

If the system design is a substantial modification of a previous design, the Systems Engineering Merit Elements should address only the engineering work associated with the modified or newly added hierarchy elements (subsystems, assemblies, or components). This includes describing how the new or modified elements were developed, how they interface with the existing system elements, and how you verified that the significant requirements identified in Systems Engineering Merit Element 2: Requirements are satisfied in the updated system.

For example, the requirements listed in Systems Engineering Merit Element 2 should pertain only to the modified or added hierarchy elements (including any related mission requirements). Likewise, the interfaces described in Systems Engineering Merit Element 3: Interfaces should address only those involving the modified or added hierarchy elements.

All eight Systems Engineering Merit Elements must follow this guidance.

1. System Hierarchy (1 point)
    a. Provide top-down breakdowns of the system design at each control gate or major review (SRR, PDR, CDR). The team may track through a single branch from an element of the SRR level hierarchy for PDR and CDR.
    b. If the team has a substantially updated system design (versus a clean-sheet-of-paper design) this year, indicate clearly on the hierarchy at each control gate or major review which hierarchy elements are from the previous design, and which are modified or new.
2. Requirements (1 point)
    a. Identify the requirements considered significant (that the team believe are most important to the mission) for system design, operations, interfaces, testing, selected engineering specialties (e.g., safety, reliability, etc.), that resulted from trade studies, etc. stated in proper "shall" language. (These significant requirements will include system and lower-level derived requirements.)
    b. Each of these requirements should specifically be addressed when the team discusses verification – see Systems Engineering Merit element 8: Verification of As-Built System Meeting Requirements.
3. Interfaces (1 point)

a. Identify the important (that the team considers important to the mission) interfaces between elements in the system hierarchy at each system hierarchy level (at a minimum, at SRR, PDR, and CDR levels).

b. Identify the type (e.g., mechanical, electrical, human, signal, data, communication, etc.) of each key important interface.

c. Identify which key important interfaces are interfaces to elements external to the system.

d. Discuss how consideration of these interfaces affected system design (how the interface constrained the design, or a design decision was based on an interface).

    a. Indicate which of the requirements listed in Systems Engineering Merit Element 2: Requirements are the result of consideration of the interfaces.

4. Engineering Specialties (1 point)

    a. Identify the engineering specialties deemed important for the system for accomplishing the mission.

    b. Discuss how considerations of these engineering specialties produced features of the system design and operations.

    c. Identify which of the requirements listed in Systems Engineering Merit Element 2: Requirements resulted from the considerations of these specialties in the design process.

    d. Indicate which of these engineering specialties played a role as evaluation criteria in trade studies.

The following are examples of a few engineering specialties that may or may not be important for the system design. There may be several others not listed here deemed important to the system. Only discuss those deemed important to the system.

- **Reliability:** what did the team have to design into the system to assure that the system will operate properly until the end of the competition.
- **Maintainability:** what had to be designed into the system to assure the maintenance and repair the system if it fails at the competition, and what tools might be needed for repairs and maintenance.
- **Logistics:** what had to be designed into the system to assure that if there is a failure at the competition that parts for repairs and maintenance will be on hand, possibly bringing spare parts or finding parts sources local to the competition.
- **Transportability:** what had to be designed into the system to be able to transport the system to and from the competition in a working condition, including possibly design features and tools needed for easy disassembly for packing/transport/shipping and reassembly at the competition.
- **Safety:** what had to be designed into the system to assure that the robot cannot cause injury or damage during the mission from pack up to leave until return home and disposal.

5. Concept of Operations (1 point)

    a. Describe how the team will operate the elements of the system hierarchy, at each system hierarchy level (at a minimum, at SRR, PDR, and CDR levels), under the environmental conditions of the competition, to accomplish the robot system mission.

   b. Indicate which of the requirements listed in Systems Engineering Merit Element 2: Requirements resulted from considerations of how the team will operate the elements of the system hierarchy to accomplish the mission in the design process.

6. Technical Performance Measurement (1 point)

A Technical Performance Measure (TPM) is any quantifiable and measurable technical characteristic that may be considered important, difficult to achieve, or particularly risky to project success, and may derive from the Project Technical Objectives. For example, it may be that total mass (high or low) is important to the team, or that a low bandwidth is difficult to achieve, or that high speed is risky for the project, or any number of other quantifiable and measurable technical quantities.

   a. Identify and discuss technical measures deemed important to accomplishing the mission and that the team used to manage the project.

   b. Provide the allocation of TPM values to elements of the system hierarchy across each level of the system hierarchy (at a minimum, at SRR, PDR, and CDR levels).

   c. Provide TPM values (estimates and final realizations after verification if possible) at least at SRR, PDR, CDR, and at the time the paper is submitted as the system design evolves through final verification.

   d. If TPM allocations across elements of the hierarchy changed at any time in the project, explain why this occurred.

   e. Demonstrate that the budgeting and management of these important technical quantities was used in management of the design process.

7. Trade Studies (1 point)

   a. Discuss how important system decisions were made using trade studies, i.e., using weighted evaluation criteria scorings of alternatives to produce a winner.

   b. Indicate which of the requirements listed in Systems Engineering Merit element 2: Requirements resulted from trade studies. (The result of any decision important enough to need a trade should by definition be captured as a significant requirement.)

8. Verification of As-Built System Meeting Requirements (1 point)

   a. Discuss how the team assured or intends to assure that the final as-built system satisfies, in the context of the concept of operations and under the environmental conditions of the competition, each of the significant requirements that the team identified in the section addressing Systems Engineering Merit Element 2: Requirements.

   b. Provide the success criterion for each verification.

19 Code of Conduct

Lunabotics is a National Aeronautics and Space Administration (NASA) Artemis Student Challenge to be conducted in a safe, respectful, and positive environment. All students and faculty are expected to act with integrity and uphold both the letter and the spirit of the rules, rubrics, and regulations.

NASA reserves the right to remove any individual or team that fails to comply with the rules, fails to follow directions of Lunabotics staff, engages in unprofessional conduct, including impersonating NASA personnel, falsely suggesting NASA affiliation or sponsorship, misusing the NASA logo, the name of a sponsoring organization, or commercial logos and trademarks.

The NASA Chief Judge and Project Manager are responsible for maintaining the integrity of the Challenge through fair and consistent interpretation, application, and enforcement of the Guidebook. All guidelines are applied equally and objectively to all participants. For matters impacting the overall Lunabotics Challenge, the Chief Judge and Project Manager will make the final determination.

20 Alignment with the National Standards in Engineering and Science

Through Lunabotics students will develop a deeper understanding of content and enhance their communication, collaboration, inquiry, problem-solving, and flexibility skills that align with the Accreditation Board for Engineering and Technology (ABET) criteria outlined below:

Criteria 3. Student Outcomes

- An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
- An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
- An ability to communicate effectively with a range of audiences
- An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
- An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives
- An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions
- An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

21 Awards

1. **Systems Engineering Paper** – Awarded to teams who demonstrate the best application of the NASA Systems Engineering process to design, build, test, and evaluate their robot (1st Place, 2nd Place, 3rd Place).

2. **Systems Engineering Leaps & Bounds Award** - Judge's recognition for significant improvement over the previous year(s) in the team's application of systems engineering to develop their robot system.

3. **Systems Engineering Nova Award** for Stellar Systems Engineering by a First Year School – Judge's recognition of team(s) who perform exceptional systems engineering in their college/university's first year in the Lunabotics Challenge as demonstrated in their systems engineering paper.

4. **The Innovation Award** – Awarded to the team with the best design based on creative construction, innovative technology, and overall architecture.

5. **The Efficient Use of Communications Award** – Awarded to the team that communicates most efficiently based on the amount of berm constructed.
6. **The Caterpillar Autonomy Award** – Awarded by Caterpillar for the most successful autonomy implementation (1st - 6th Places).
7. **The Construction Award** – Awarded to the teams that score the most points during berm building operations in the Artemis Arena (1st Place, 2nd Place, 3rd Place).
8. **The Artemis Grand Prize** – awarded to the team that completes all events and scores the most points per section 17.2 Artemis Grand Prize Score.

Note: The Awards list is not all-inclusive; and is subject to change without notice.

22 References

22.1 NASA Infrastructure Pilot Excavator (IPEx)

The primary goal of NASA's Infrastructure Pilot Excavator (IPEx) project is to dig up lunar soil and transport it across the Moon's surface. This process is designed to enable the extraction of vital resources, such as hydrogen, oxygen, and water, which are essential for life support systems. IPEx is a groundbreaking robotic system designed by NASA to revolutionize lunar excavation. Functioning as both a bulldozer and a dump truck, IPEx is engineered to efficiently mine and transport lunar regolith, the loose rocky material on the Moon's surface, which is crucial for future lunar missions and In-Situ Resource Utilization (ISRU) processes. This dual capability makes IPEx an indispensable tool for sustainable lunar exploration. Technologies that enable affordable, autonomous manufacturing or construction, as well as long-term system maintenance, allow a variety of missions to be simultaneously carried out by NASA and its partners.

22.2 Resources Available Project Management and Systems Engineering

Project Management and Systems Engineering educational videos can be viewed at **NASA Lunabotics Videos**. These videos include a series titled ***Systems Engineering for University Level Engineering Competitions Videos*** that introduces key project management and systems engineering concepts and methods useful for the Lunabotics Challenge.

Also included are **seven** different project management and systems engineering **seminars** that were recorded live at Lunabotics Challenges at Kennedy Space Center. The most immediately relevant of these is the seminar titled ***How to Develop the Perfect Lunabotics Systems Engineering Paper: And Win the Whole Competition***. We recommend viewing this video at the very start of the project.

22.3 Robot Mobility in Regolith Simulant

Robot mobility has been a common issue for teams that have never run their robot in Lunar regolith simulant. As such, it is strongly recommended to review the following Mobility Workshop: Dr. Philip Metzger on YouTube.

22.4 See past Lunabotics Challenges:

https://www.youtube.com/watch?v=b0iyM1ecZR0

https://www.youtube.com/watch?v=nWq2yC9RRvg

https://www.youtube.com/watch?v=cuZEIy46nzY

22.5 Frequently Asked Questions (FAQ's)

The Guidebook and FAQs are complementary documents; teams are required to review both. FAQs will be located at the AMF Lunabotics Challenge website. We can respond to FAQ's as follows:

1. From the Team Lead.
2. From an authorized email address, ending in ".edu".
3. Can only address inquiries from the current Guidebook.
4. Use the format: School Name, Rule Paragraph Number & Question